

Improving convergence in SPH

Robert Wissing - University of Oslo

Co-funded by the European Union

Funded by the European Union. This work has received funding from the European High Performance Computing Joint Undertaking (JU) and Belgium, Czech Republic, France, Germany, Greece, Italy, Norway, and Spain under grant agreement No 101093441.

EuroHPC Joint Undertaking

What is SPACE ?

Stands for scalable parallel astrophysical codes for exascale

Newly funded Centre of Excellence (CoE), which aims to prepare astrophysics applications for the exa-scale and beyond.

Funded by the Eurohpc joint undertaking.

Brings together expertise from scientists, code developers, software developers and hpc experts



Activities in SPACE related to ChaNGa

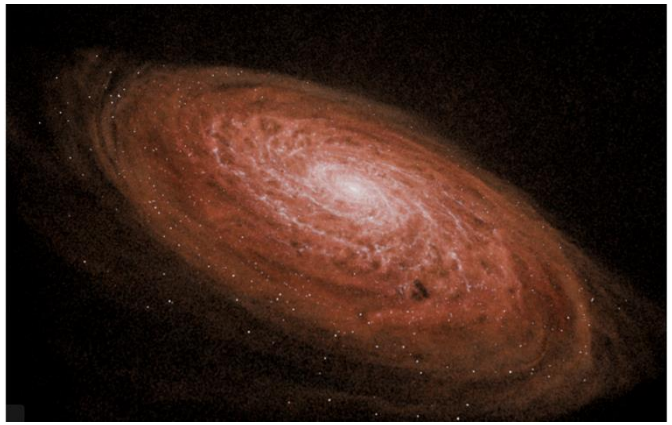
Objectives for ChaNGa within SPACE:

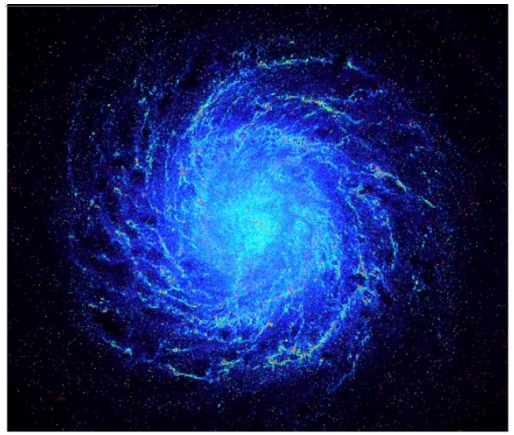
Improve scaling and convergence of the ChaNGa code

Support in development of GPU version

Generate an extensive test suite and CI/CD (github.com/robertwissing/testsuite)

Port and scale on all European JU systems





Performance analysis and co-design

Performance profiling and benchmarking

Evaluation of energy efficiency

HBM vs DDR study

Profiling on vectorization

Visualization and analysis

High-performance scalable pipeline using Blender

Cinematic visualization in Houdini

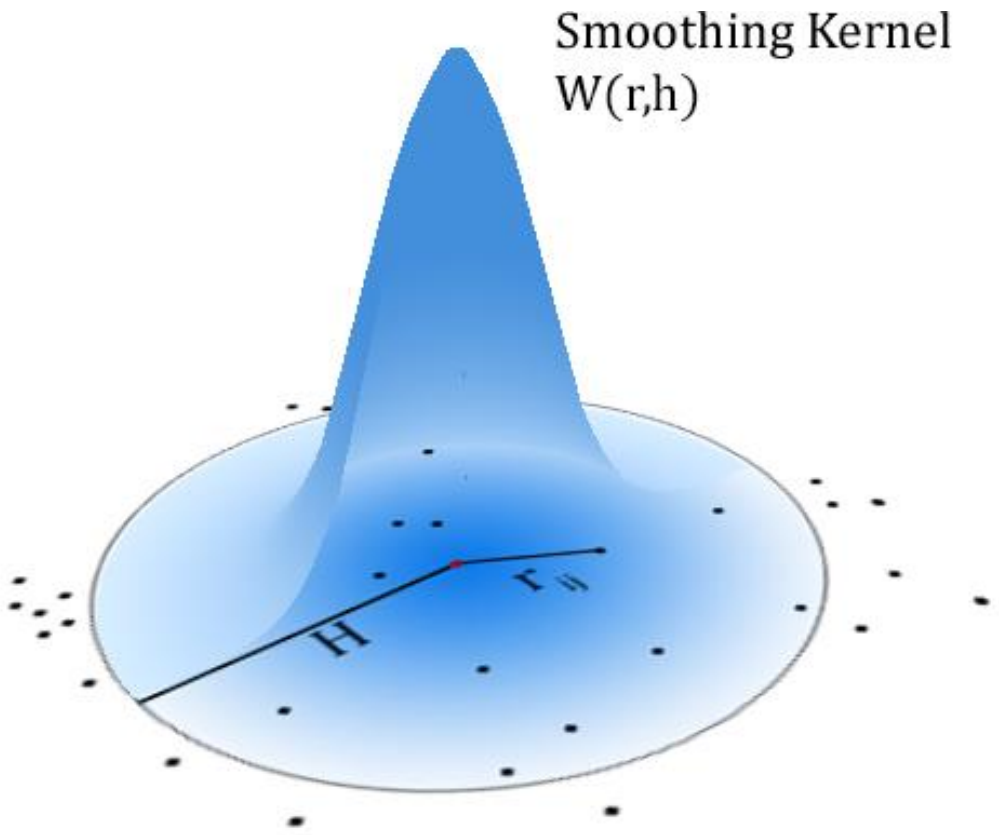
In-situ Parallel visualization through Hecuba

Machine learning

Representation learning for explorative knowledge discovery

Predicting the effect of radiation on gas cooling rates on-the-fly

Smoothed particle hydrodynamics (SPH)



Smoothing length (h) represents our resolution scale. Similar to Δx in grid

- Lagrangian particle method
- Describes the continuous fluid at any point by interpolating over particles (interpolation points)
- Uses a **Kernel function** W for interpolation.
- Compact support

$$W(r,h) = 0 \; when \; r > H$$

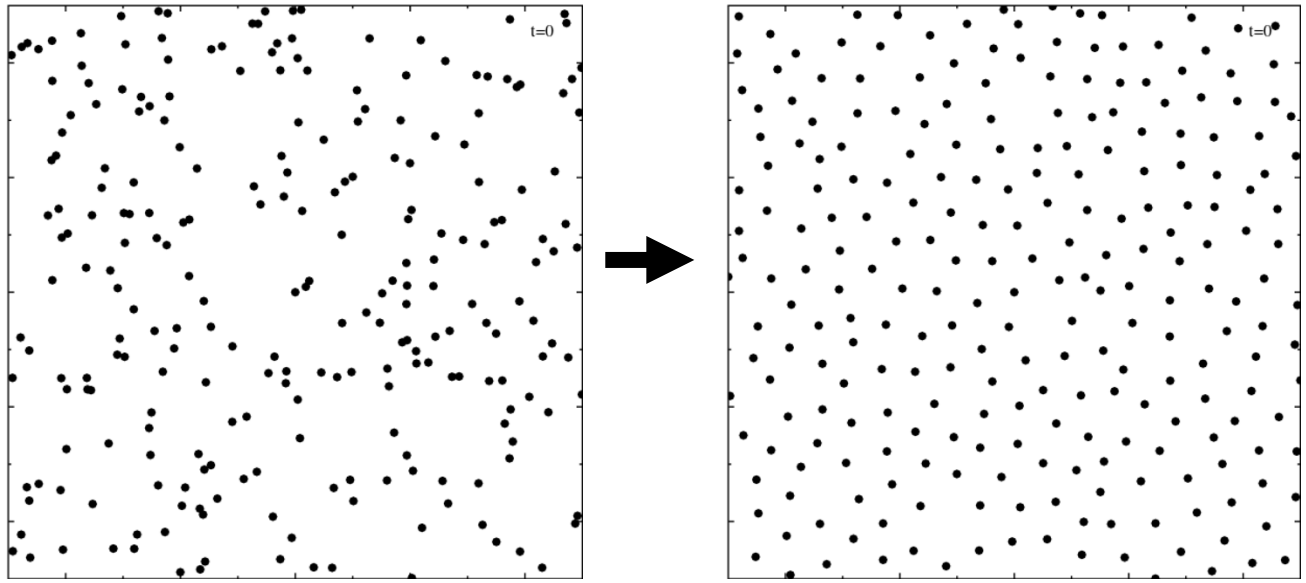
The low order errors of SPH

- SPH is second order if the particle distribution is well ordered, but has both zeroth and first order errors when particle distribution is unordered.
- That is even if pressure is constant, particles can still move in the aim to regularize the particle distribution

$$P = constant$$

$$\frac{dv}{dt} = -\frac{\nabla P}{\rho_a} \neq 0$$

$$\frac{dv}{dt} = -\frac{\nabla P}{\rho_a} \approx 0$$



The low order errors of SPH

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- That is even if pressure is constant, particles can still move in the aim to regularize the particle distribution
- Low order errors are deeply related to the effective flux areas of particles by two relations:

Cancellation of areas

$$\int_S \hat{n} dS = 0$$

Consistency between area and volume

$$\frac{1}{3}\int_S (\mathrm{r} \cdot \hat{n}) dS = V$$

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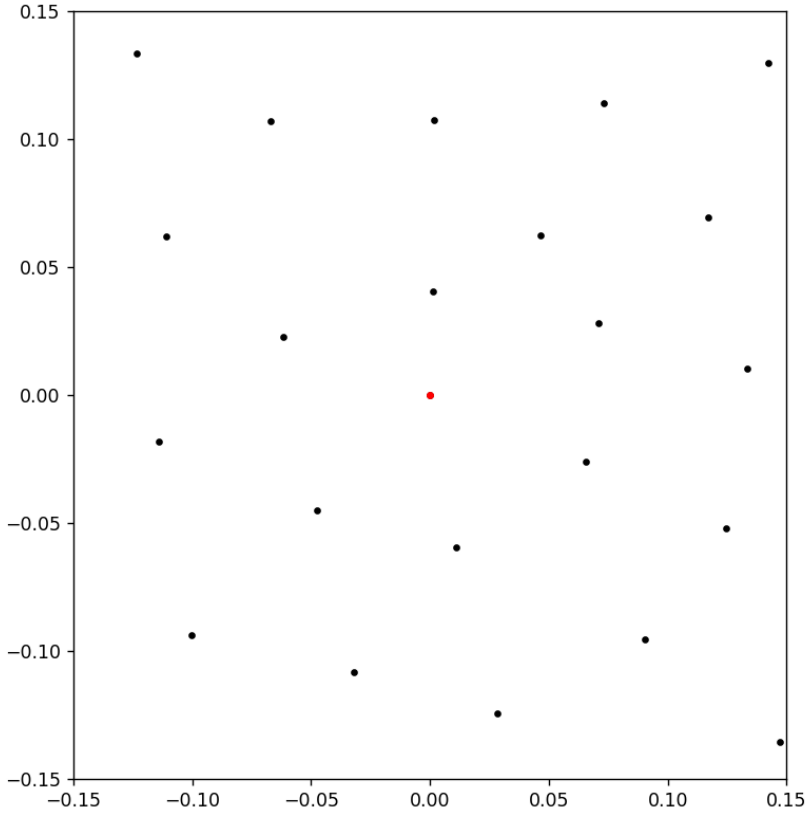
$$\frac{1}{3}\int_S (\mathrm{r} \cdot \hat{n}) dS = V$$

Effective flux area in SPH

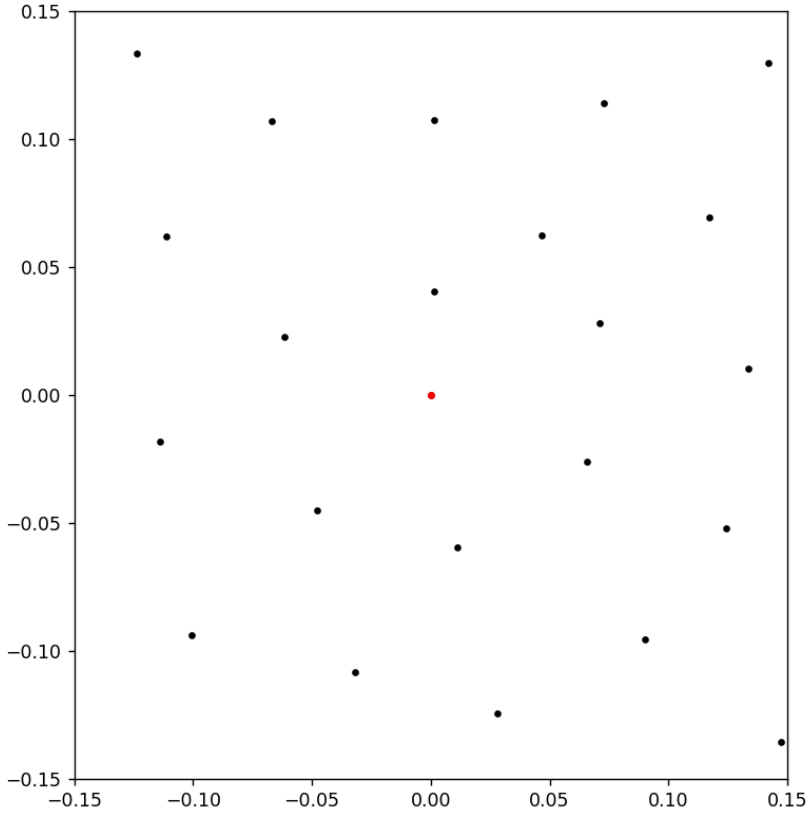
$$A_{ab} \propto \mathrm{V_a V_b} \nabla W_{ab}$$

The low order errors of SPH

SPH



Voronoi



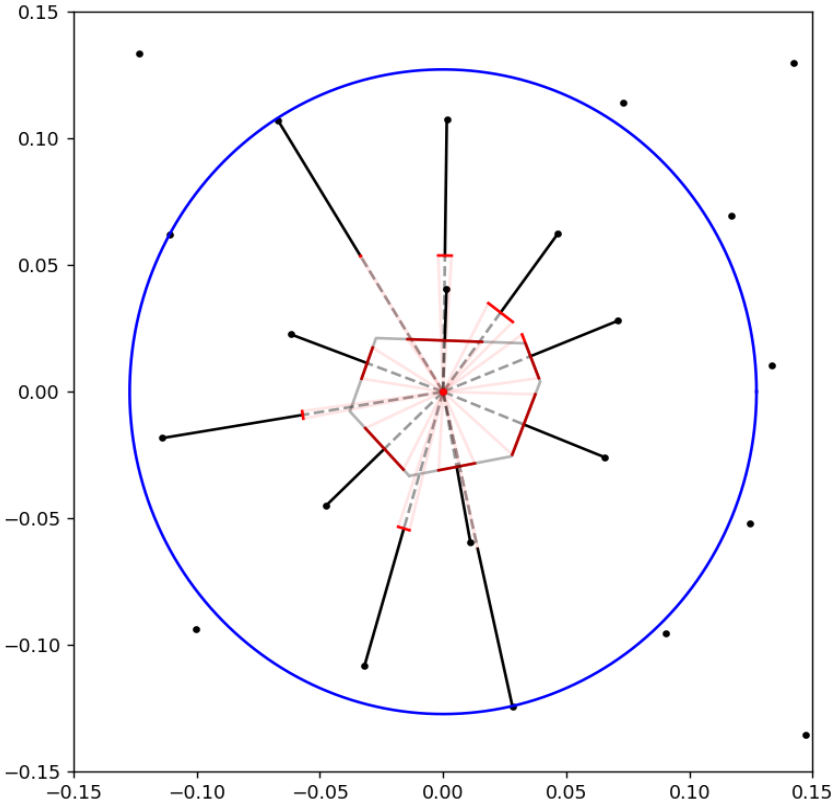
Initial fluid elements locations

(Owens)

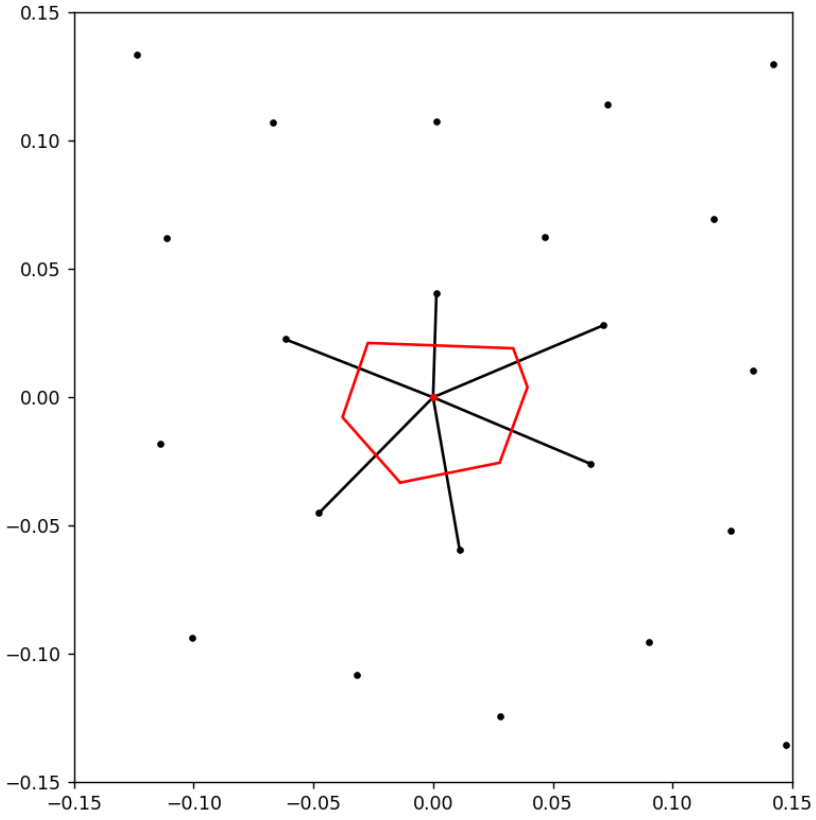
The low order errors of SPH

SPH

$$\int_S \hat{n}\, dS \neq 0$$



Voronoi



Effective flux areas for the red particle

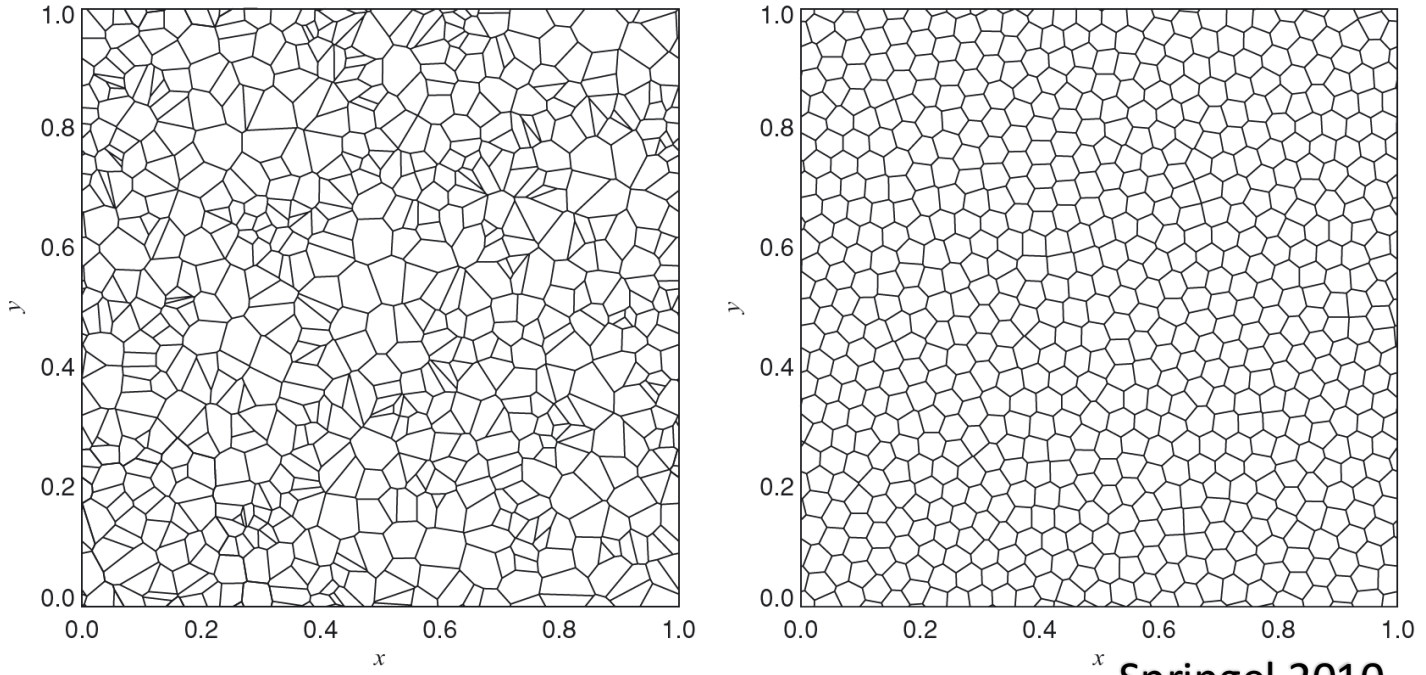
$$\int_S \hat{n}\, dS = 0$$

(Owens)

The low order errors of SPH

- Regularization of resolution elements is not unique to SPH, also done in moving mesh to keep the mesh well-shaped.

Note: Not gradient error!
Done to minimize other errors (reconstruction etc.)



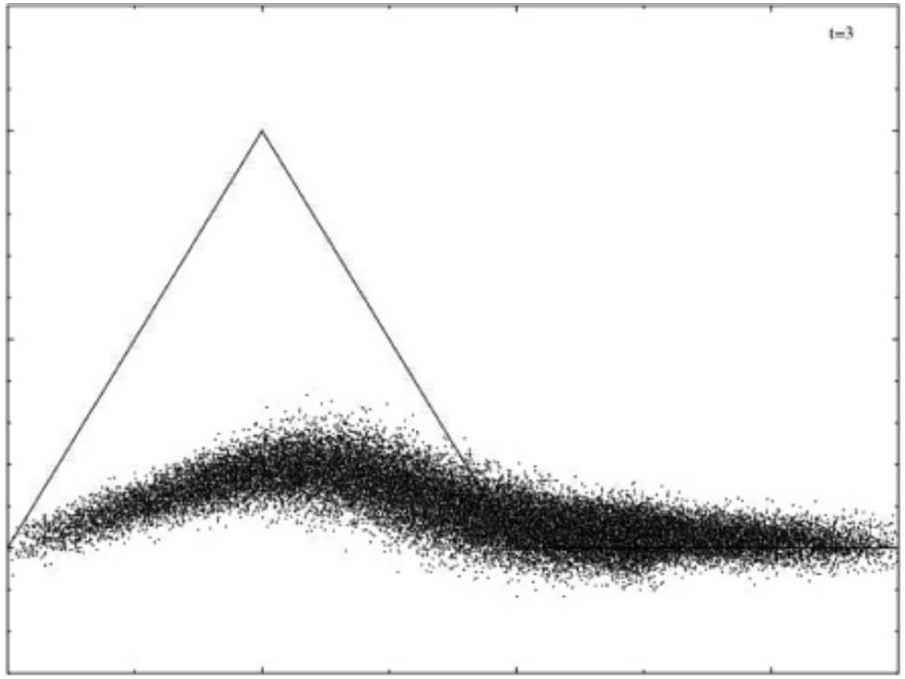
Springel 2010

The low order errors of SPH

- Regularization of resolution elements is not unique to SPH, also done in moving mesh to keep the mesh well-shaped
- The low order errors of SPH are problematic in:

1. Sub-sonic flows: where the ”regularization” noise become comparable to the dynamics

Gresho-Chan vortex

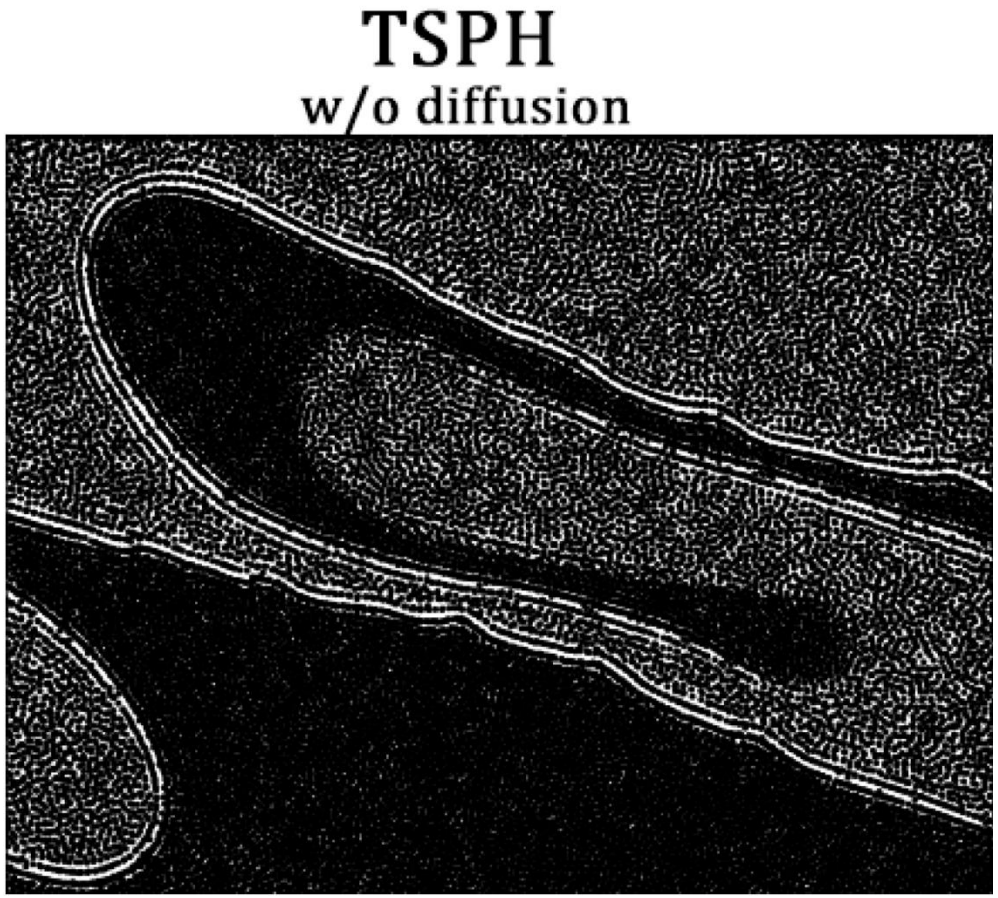


$\mathcal{M} \sim 0.3$

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- The low order errors of SPH are problematic in:

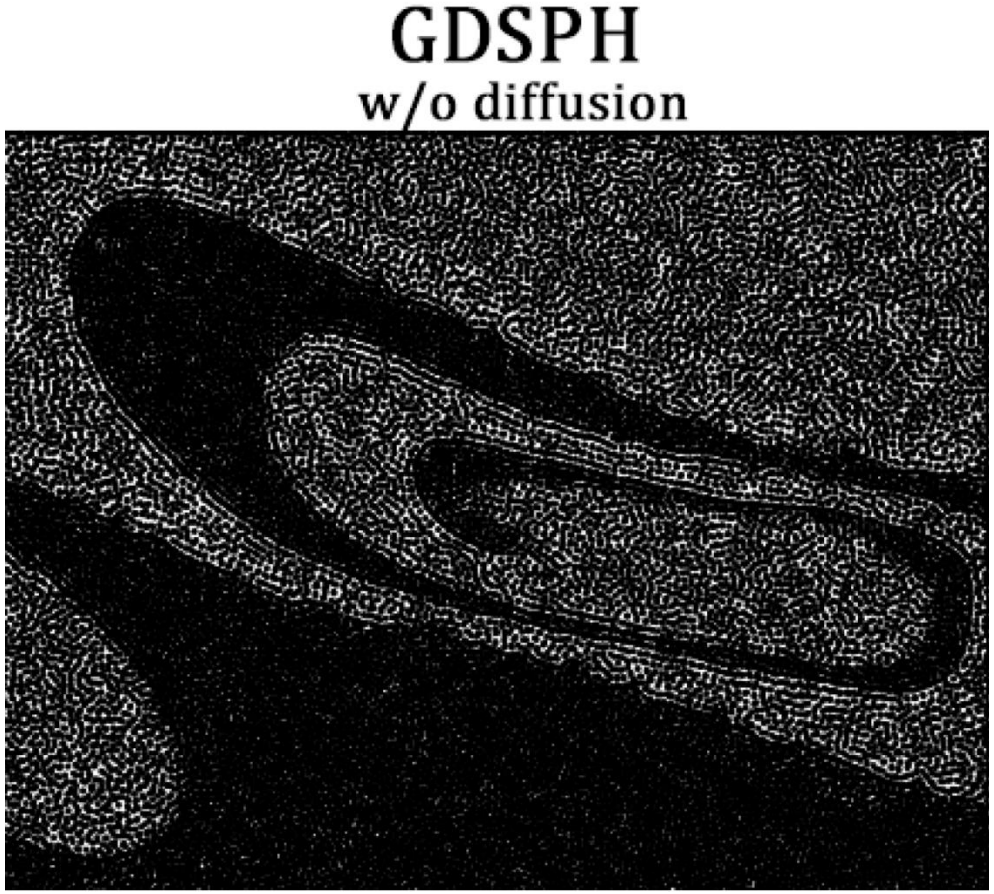
1. Sub-sonic flows: where the ”regularization” noise become comparable to the dynamics
2. Density discontinouties: where no ”good” partitioning of particles exist



Improving the low order errors

- We can improve the low order errors in SPH by:

1. Choice of volume weightings in our gradient operator TSPH→GDSPH (Gasoline2)
2. Apply corrections to the gradient and kernel (ISPH,GIZMO, etc.)
3. Choice of smoothing kernel and number of neighbours
4. Dissipation scheme
5. Other choice of particle volume/density (DISPH, PSPH, etc.)



Linear-exact gradients in SPH

- We can effectively remove our first order errors by correcting our gradients

- Several flavors of this:

1. Simple inversion of first-order error term
   (Bonet 1999, Price and Monaghan 2004)
2. ISPH - inversion with Brookshaw method for first derivatives ($\nabla W_{ab} = -r_{ab} W_{ab}$)
   (Garcia-Senz 2012)
3. Linear-order reproducing kernels
   (Frontiere 2017, Rosswog 2025)

ISPH

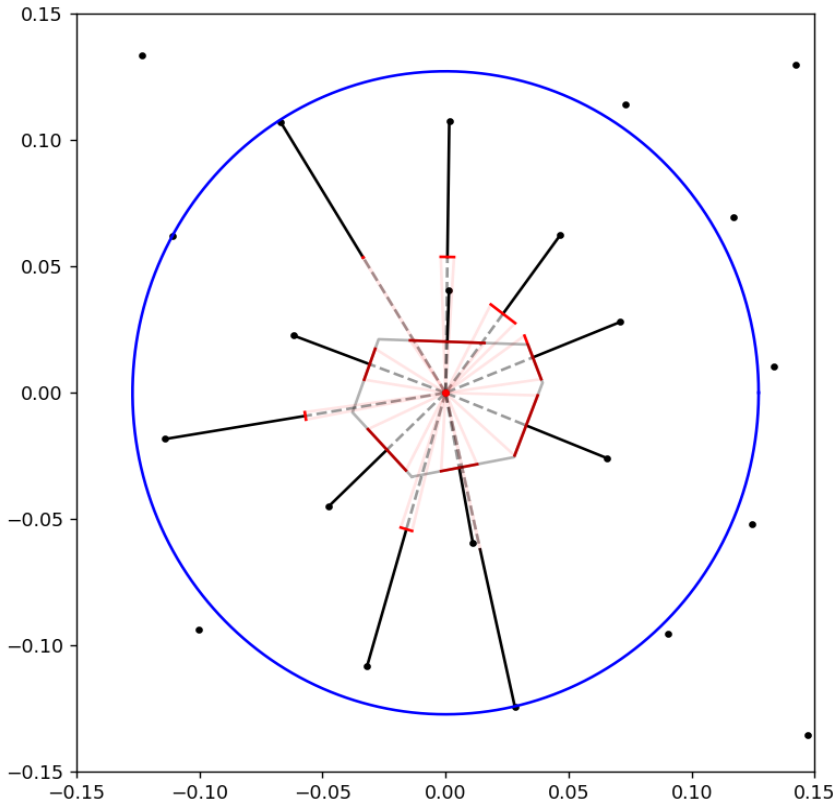
$$G_a^k = \sum_1^3 C^{kd} r_{ba}^d W_{ab}$$

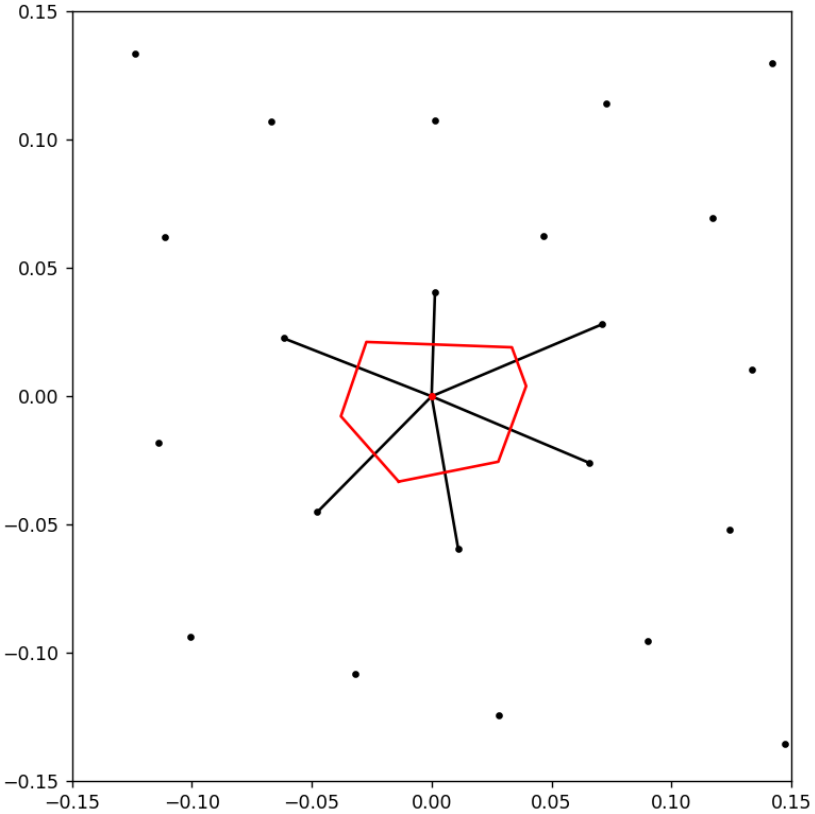
$$C^{ki} = \left(\sum_b V_b r_{ba}^k \, r_{ba}^i W_{ab} \right)^{-1}$$

We effectively replace $\nabla W_{ab} \rightarrow G_a^k$ in our equations

Linear-exact gradients in SPH (ISPH)

SPH



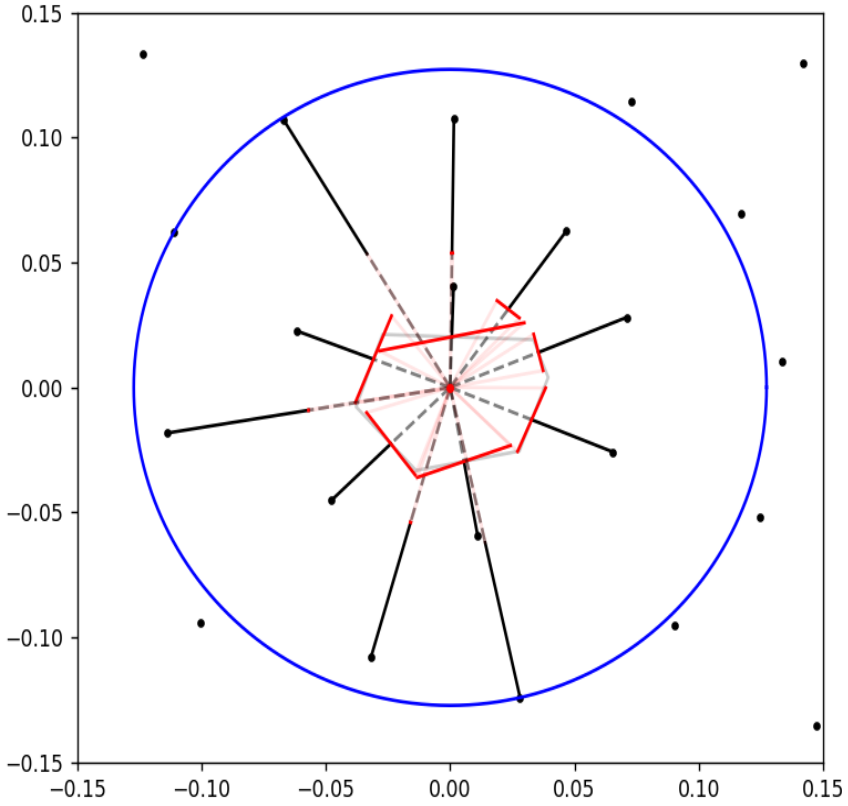


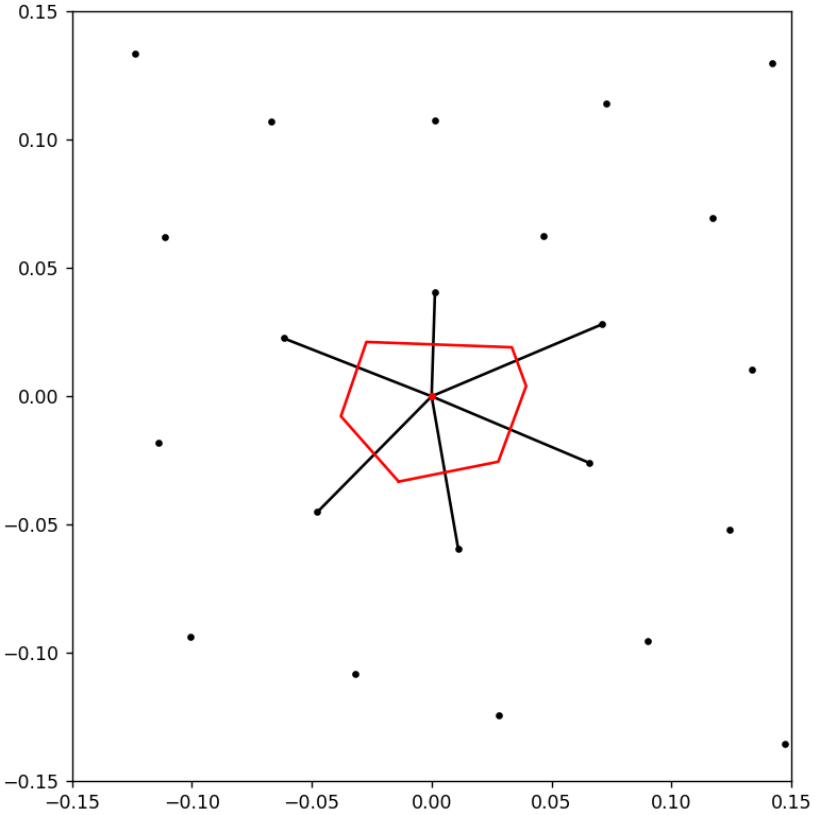
Voronoi

(Owens)

Linear-exact gradients in SPH (ISPH)

ISPH





Voronoi

$$A_{ab} = \mathrm{V_a V_b} \overline{G_a}$$

Ensures consistency between area and volume

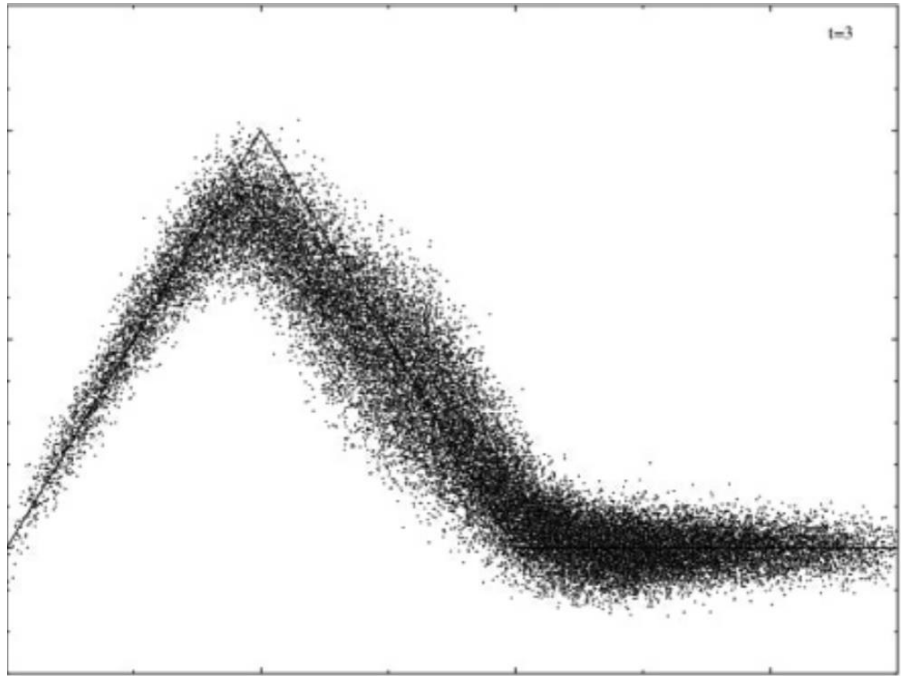
$$V_a = \frac{1}{3}\sum_b A_{ab} \cdot r_{ab} = \frac{V_a}{3}\sum_b V_b (r_{ab} \cdot \overline{G_a})$$

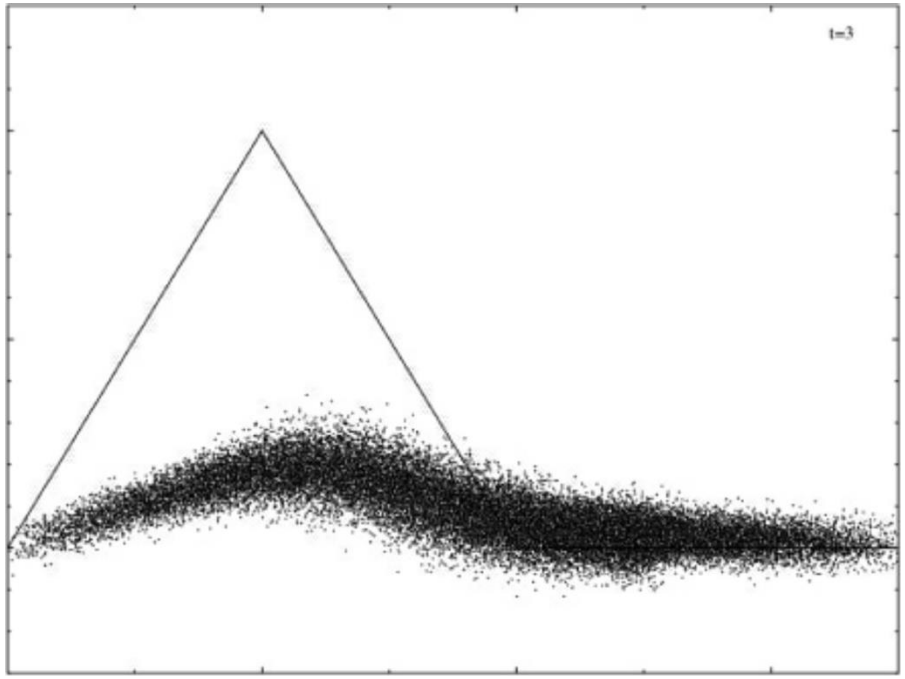
Improvements in sub-sonic flows

ISPH

GDSPH

$t = 3$





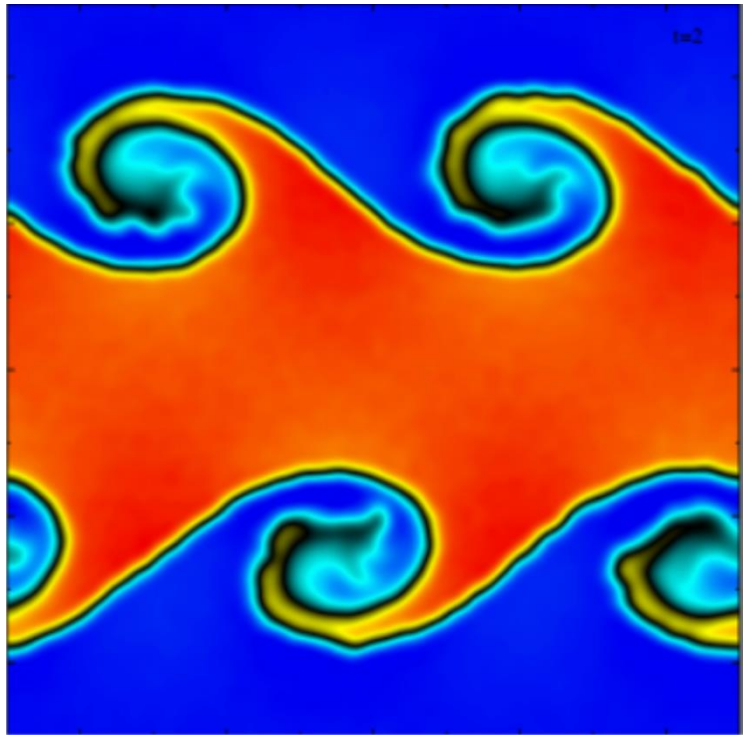
$n_x = 64. N_{smooth} = 64$ Monaghan AV ($\alpha = 0.05$ $\beta = 2.0$)

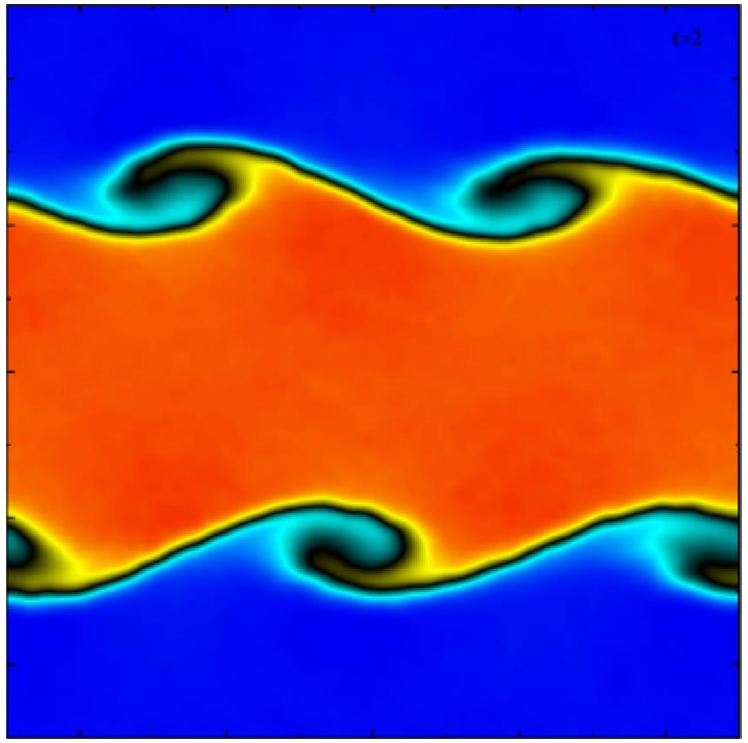
Improvements in sub-sonic flows

ISPH

GDSPH

$t = 2$





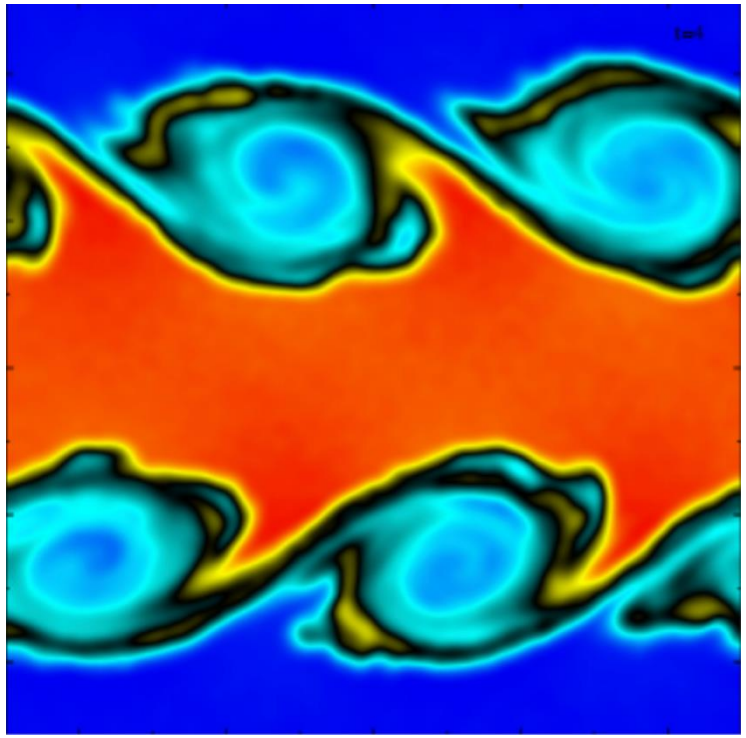
$n_x = 128. N_{smooth} = 64$ Monaghan AV ($\alpha = 0.05$ $\beta = 2.0$)

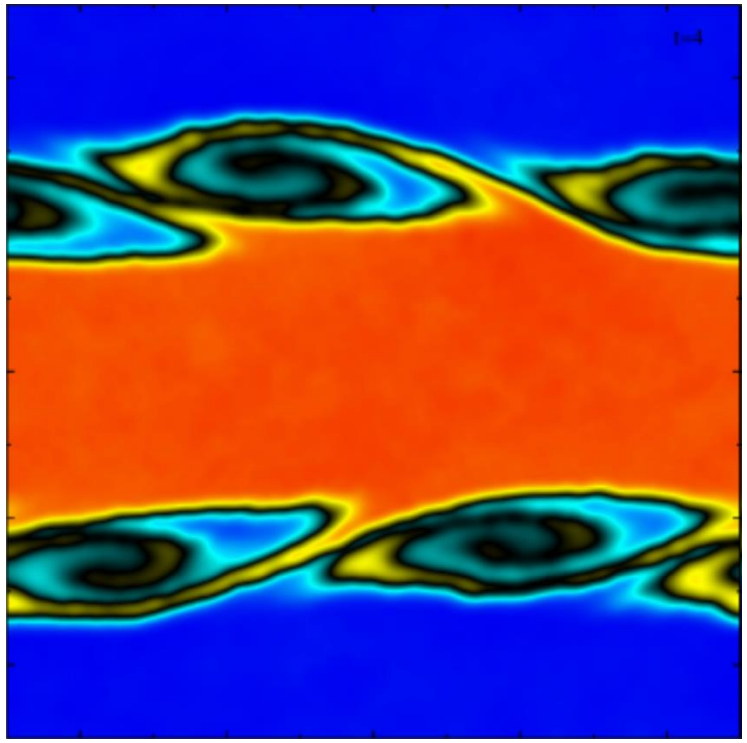
Improvements in sub-sonic flows

ISPH

GDSPH

$t = 4$

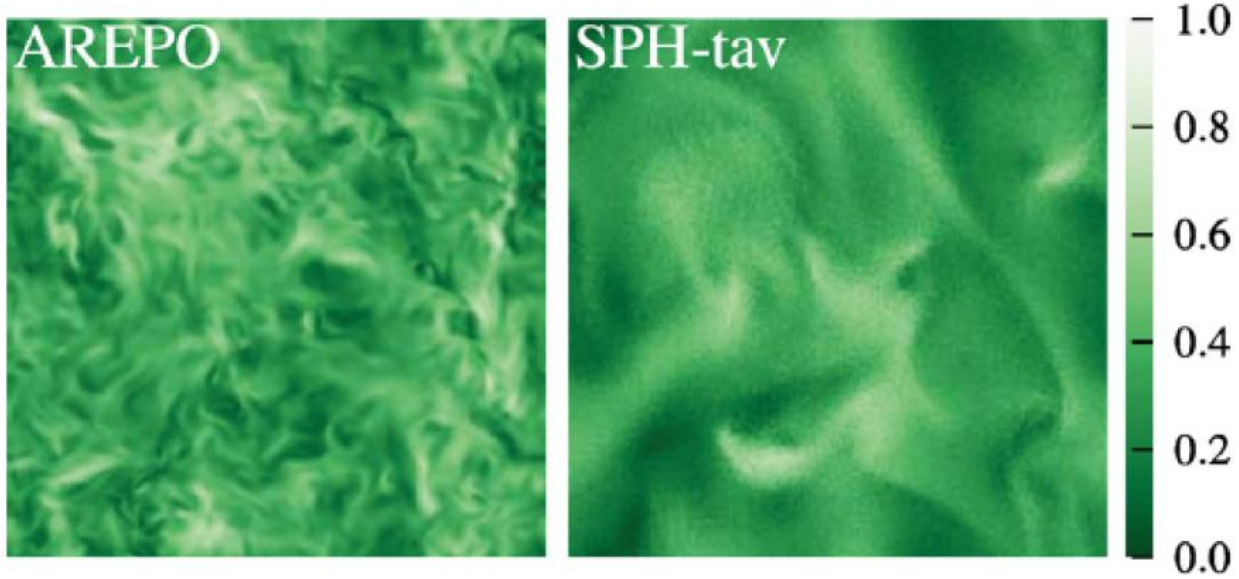




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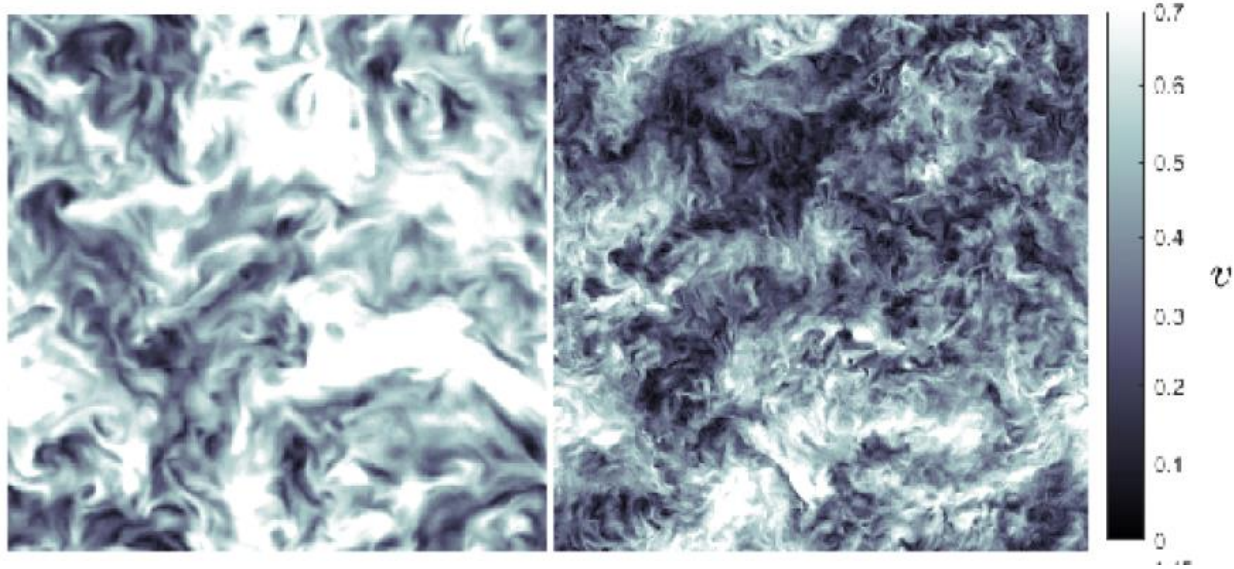
Improvements in sub-sonic flows

$\mathcal{M} \sim 0.3$



SPH-EXA (ISPH)
$N = 250^3$

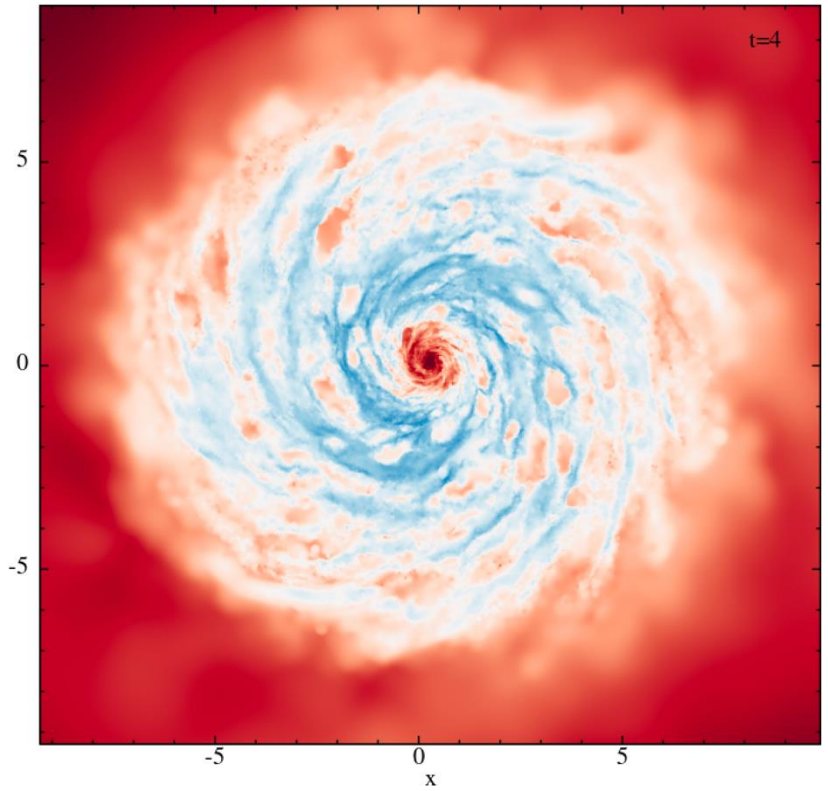
SPH-EXA (ISPH)
$N = 1000^3$



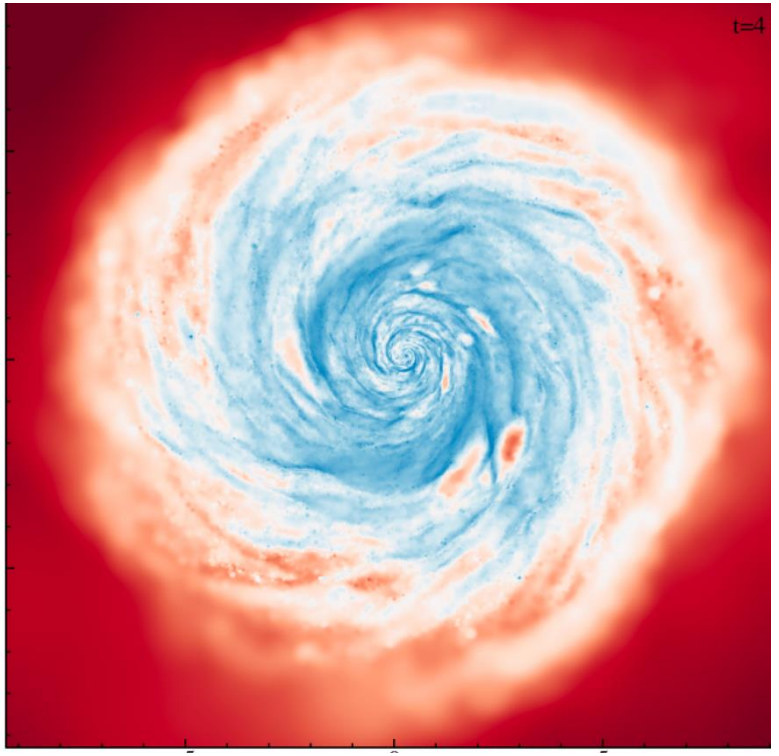
Cabezon et al. 2025

Important for magnetic field in galaxies?

GDSPH



ISPH

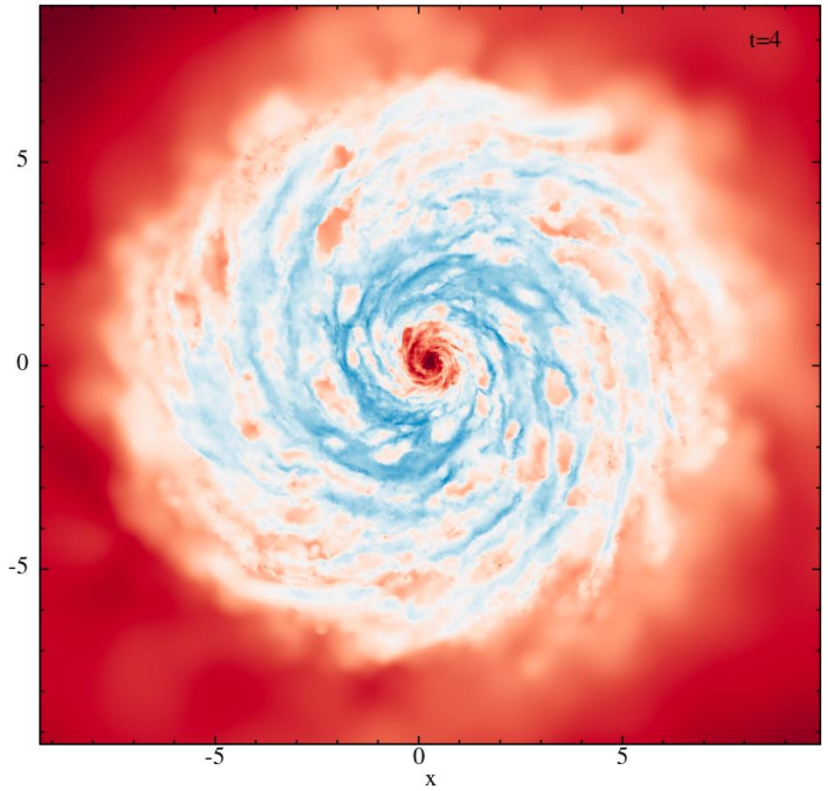


Rendering shows B^2
Blue strong
Red weak

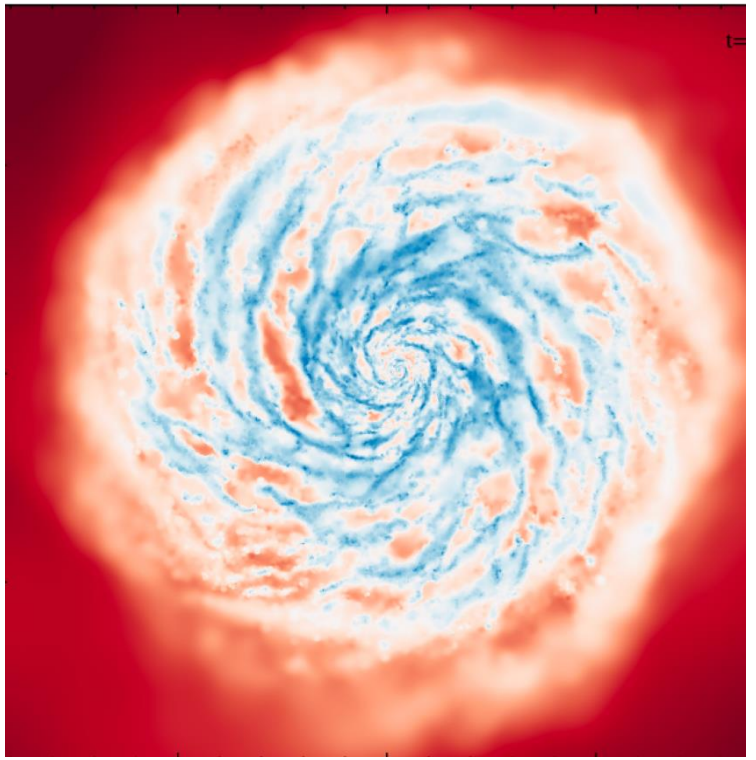
$N_{Jeans} = 8$

Important for magnetic field in galaxies?

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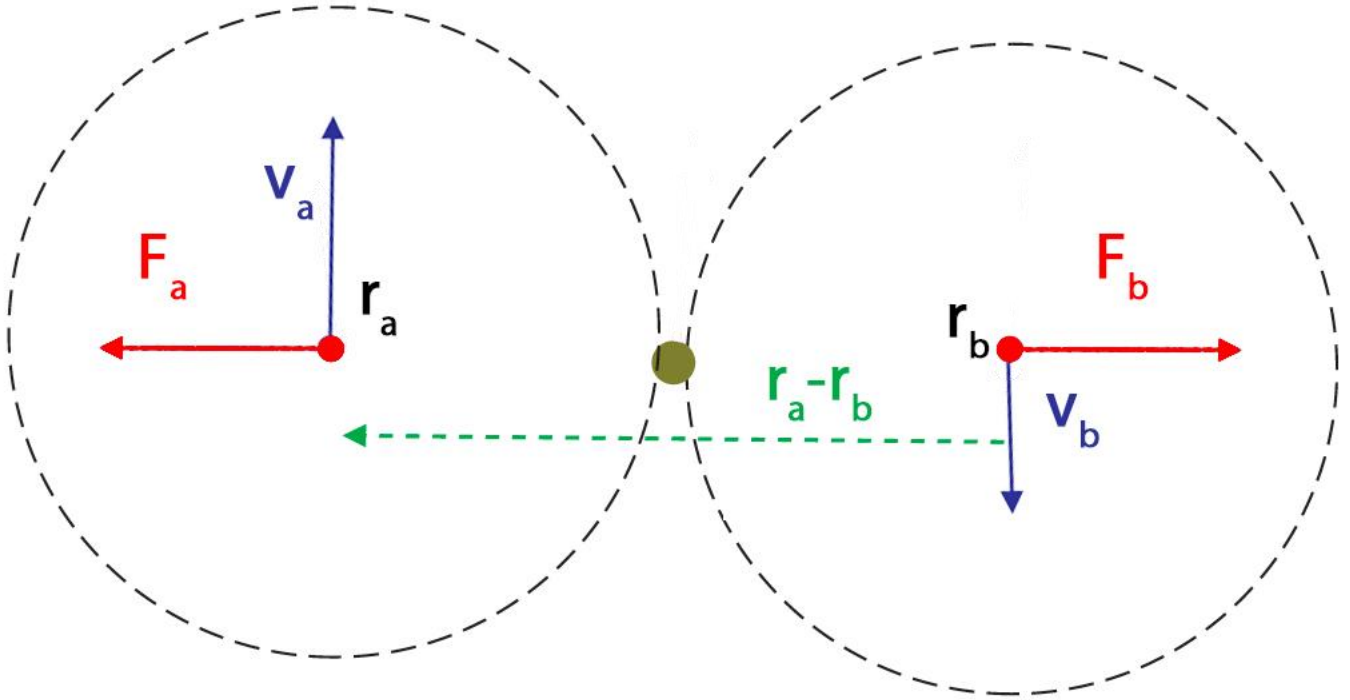
Rendering shows B^2
Blue strong
Red weak

$$N_{Jeans} = 4$$

Issues with linear-exact gradients ?

- While local angular momentum is often improved due to better gradients, global angular momentum is not ensured to be conserved
- Similar issues when one adds anisotropic forces to SPH (magnetic field tension, physical viscosity, strength, etc.) or doing particle merging

Isotropic force

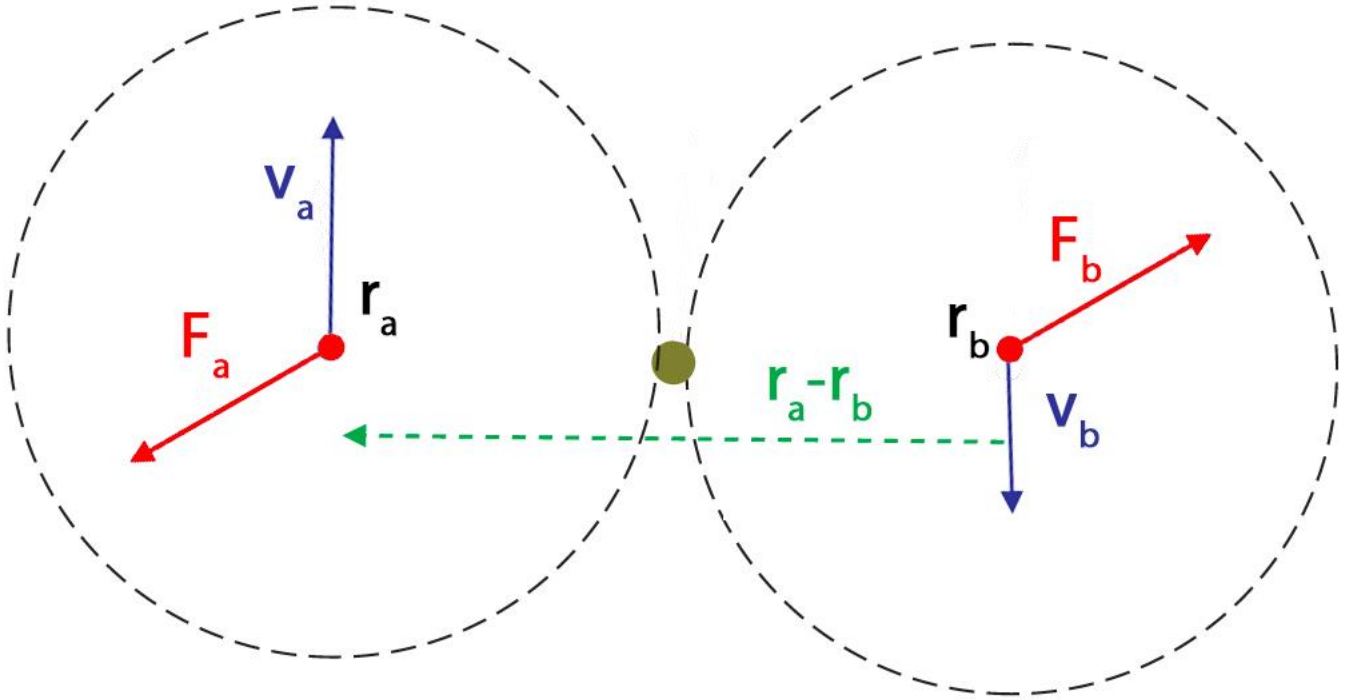


$$\Delta L_{ab} = 0$$

Issues with linear-exact gradients ?

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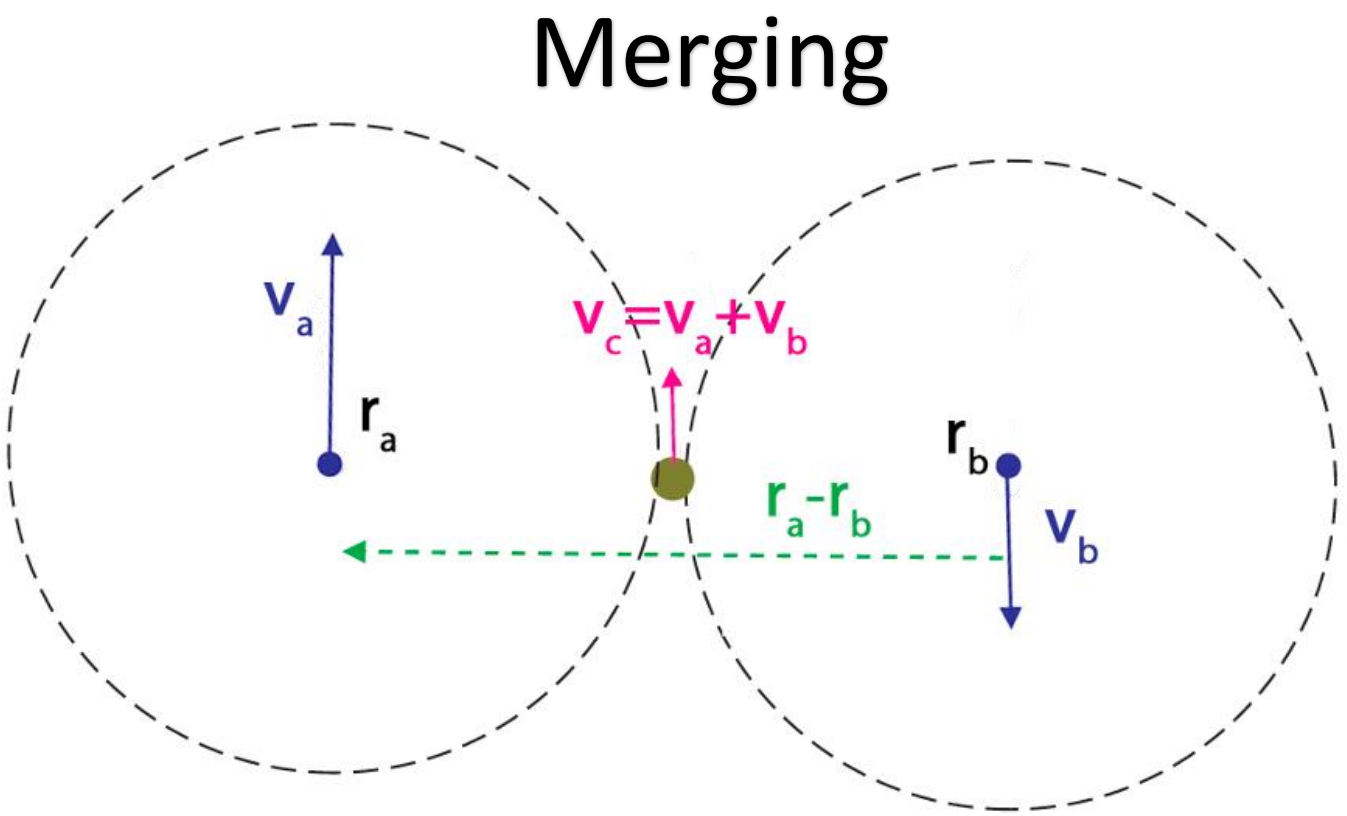
Anisotropic force / ISPH



$$\Delta L_{ab} \neq 0$$

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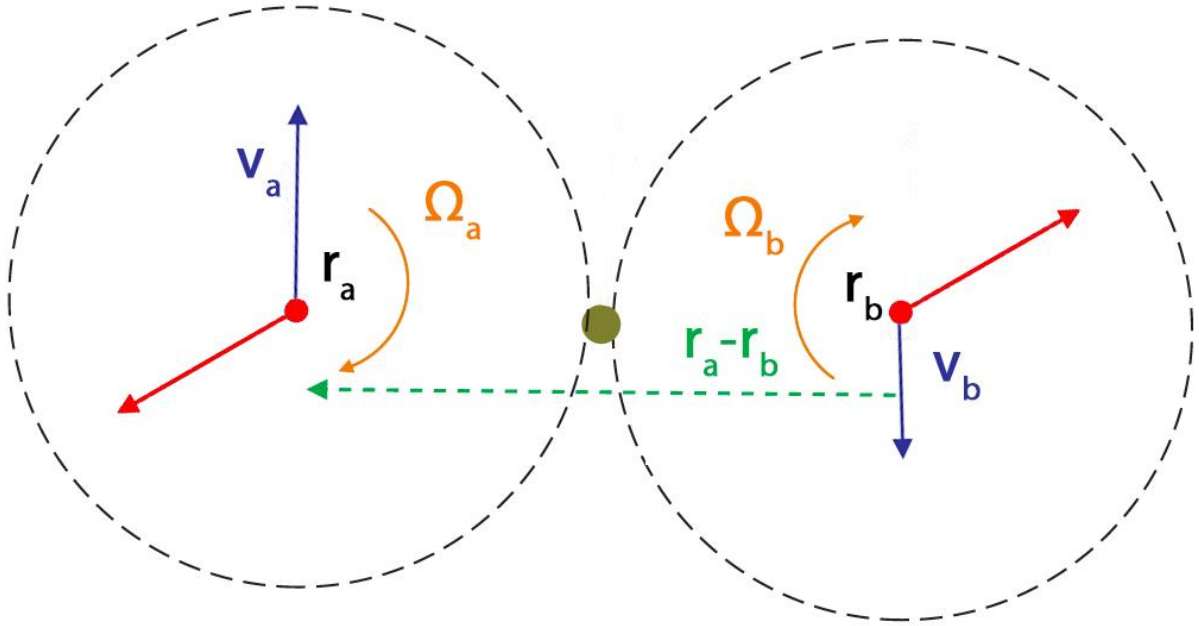
$$if\ v_a = -v_b.\quad L_c = 0\ E_k = 0$$

Spin algorithm

- A way to restore global angular momentum is through particle spin/unresolved angular momentum

$$m_a \frac{dv_a}{dt} = \sum_b F_{ab}$$

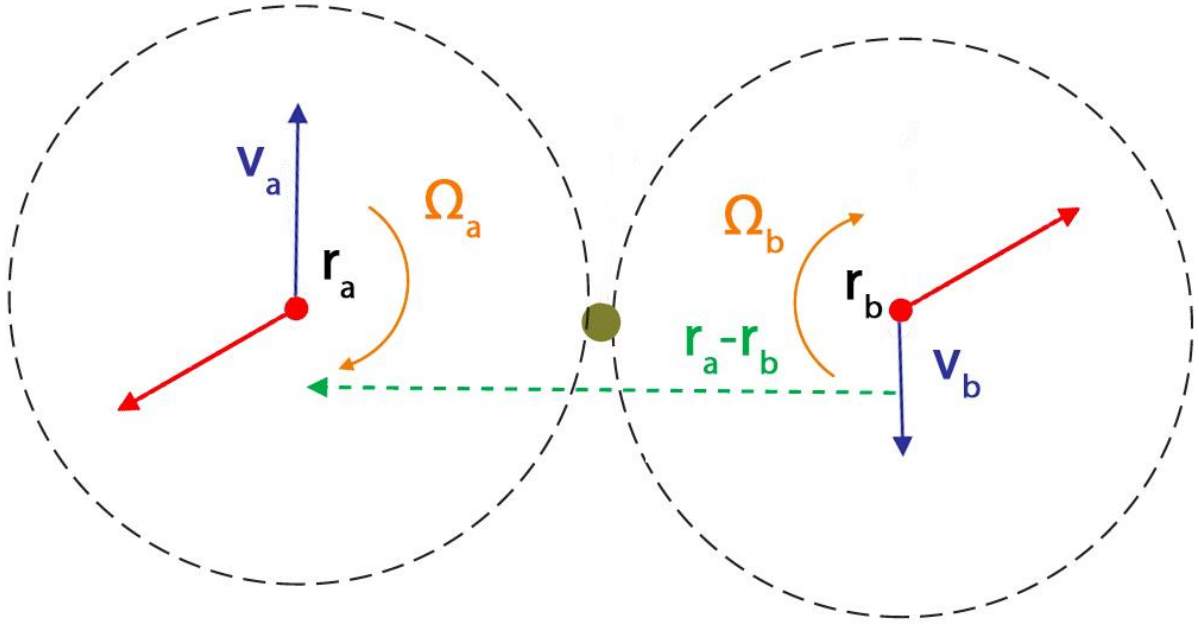
$$I_a \frac{d\Omega_a}{dt} = \frac{1}{2} \sum_b r_{ab} \times F_{ab}$$



Spin algorithm

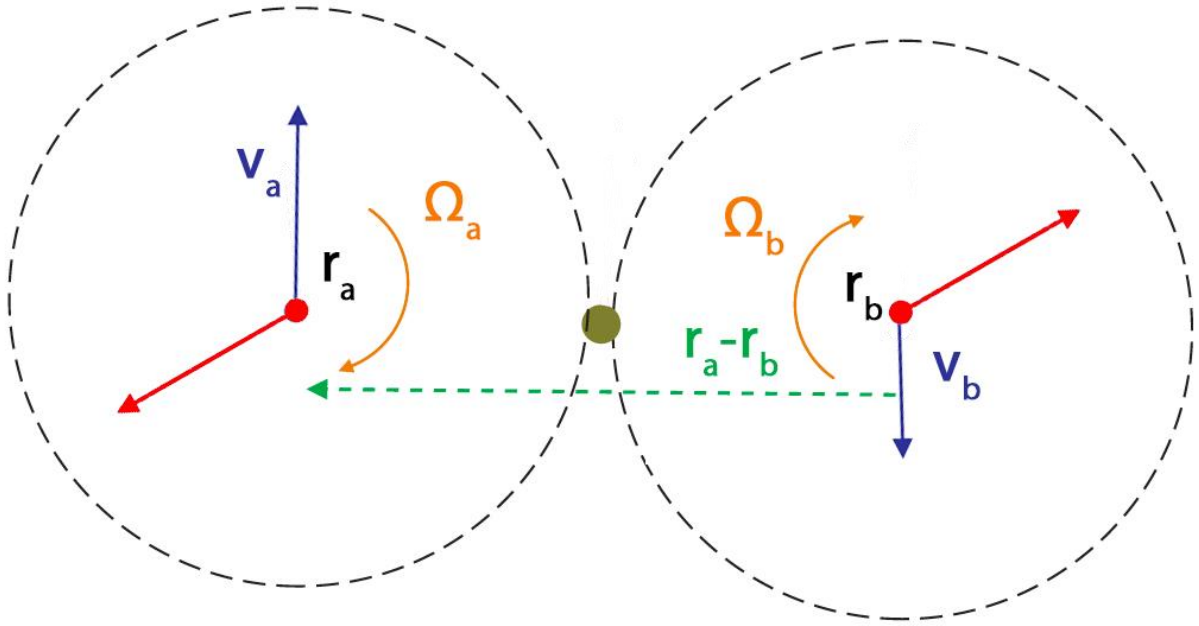
- The challenge is how to reintroduce the angular momentum to the resolved fluid
- We can take inspiration from micrpolar fluid SPH (Souto-Iglesias 2021) to construct a numerical ”spin” force

$$F_{\Omega,num} = 2\eta_r \nabla \times L_\Omega$$



Spin algorithm

- **Numerically induced angular momentum errors**:
- There is no physical energy in the spin
- Energy transfer occur between resolved kinetic and thermal energy.
- Need to adhere to positive entropy contribution.
- Corrections to angular momentum is only allowed to be done by a decrease in kinetic energy.
- Acts as a viscosity to regain angular momentum conservation to the fluid

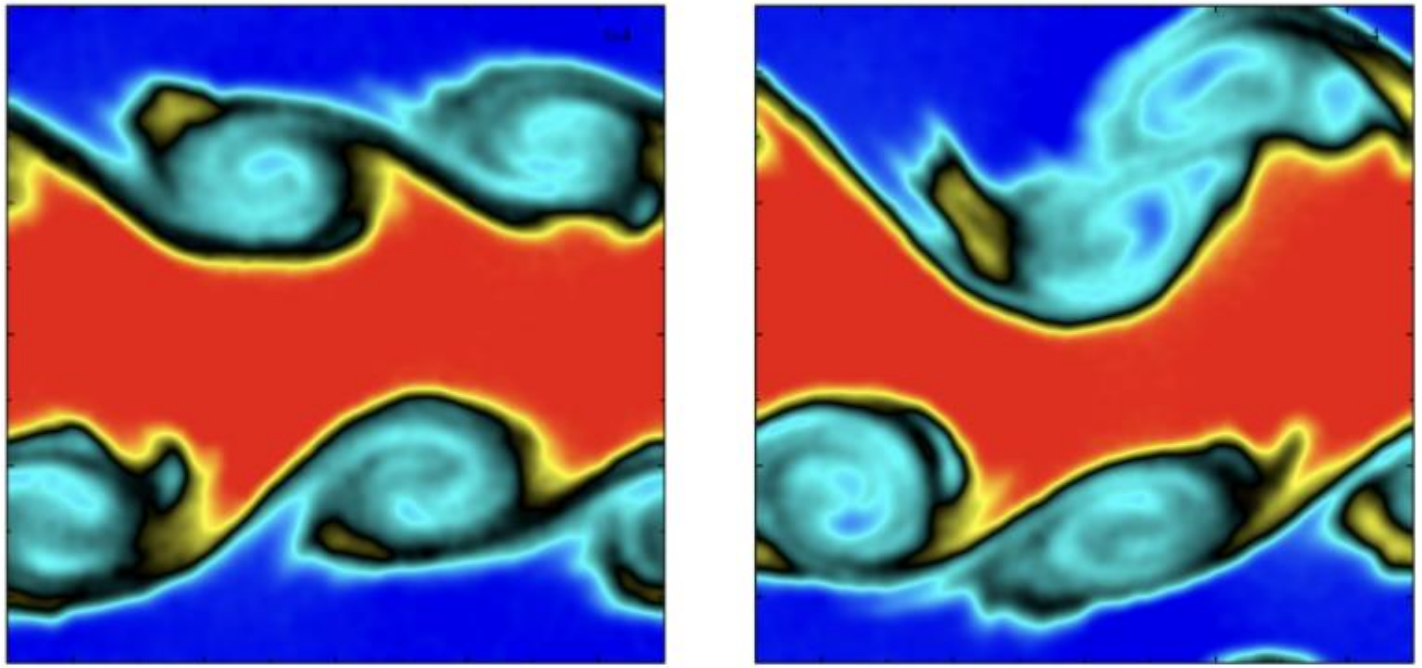


Force is adjusted such that it follows the hydro timestep and thus does not cost any additional computation

Results of using spin algorithm

ISPH $N_{smooth} = 32$ Wendland C4 -> Very Noisy!
Induces a lot of angular momentum errors for ISPH

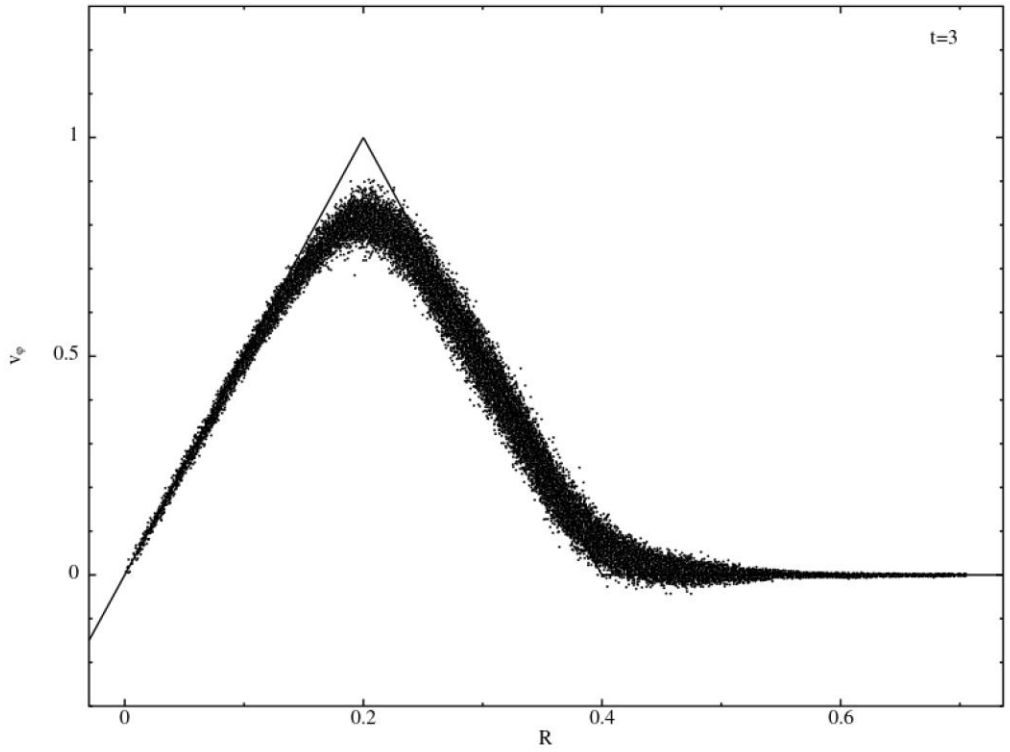
Spin



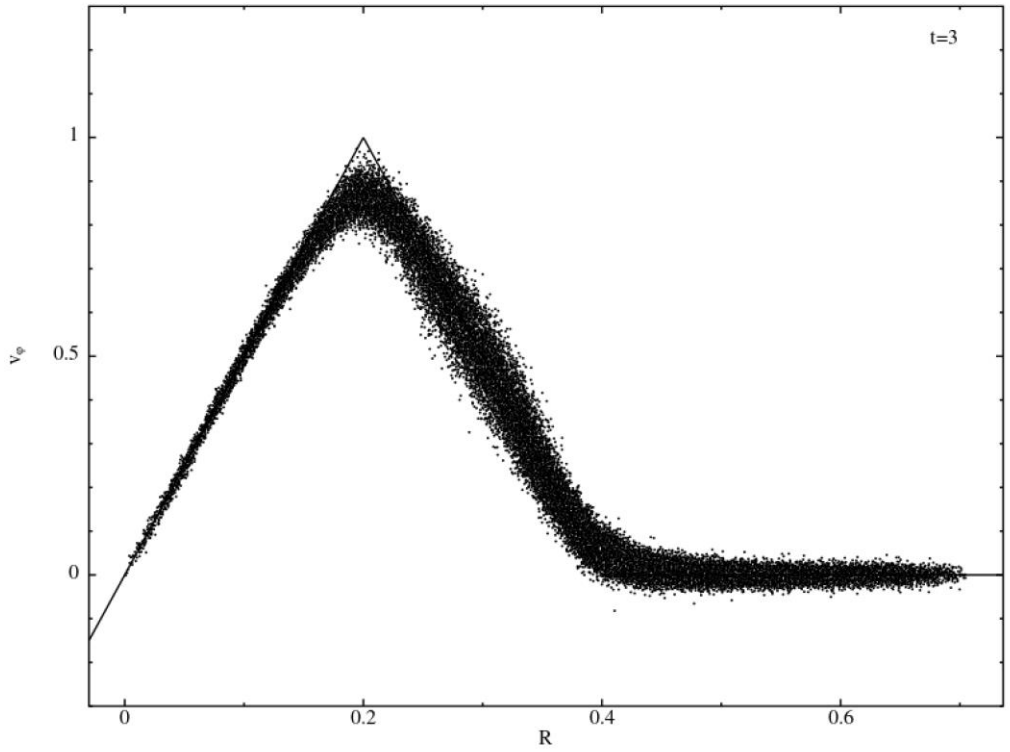
No Spin

Results of using spin algorithm

Spin



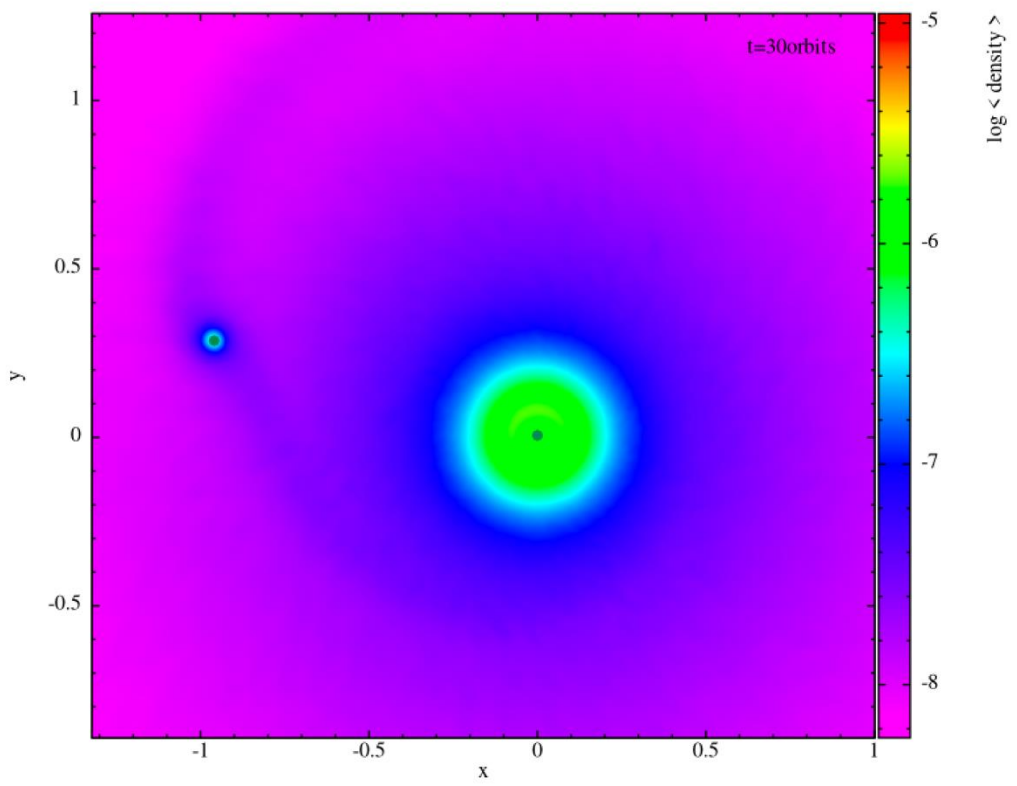
No Spin



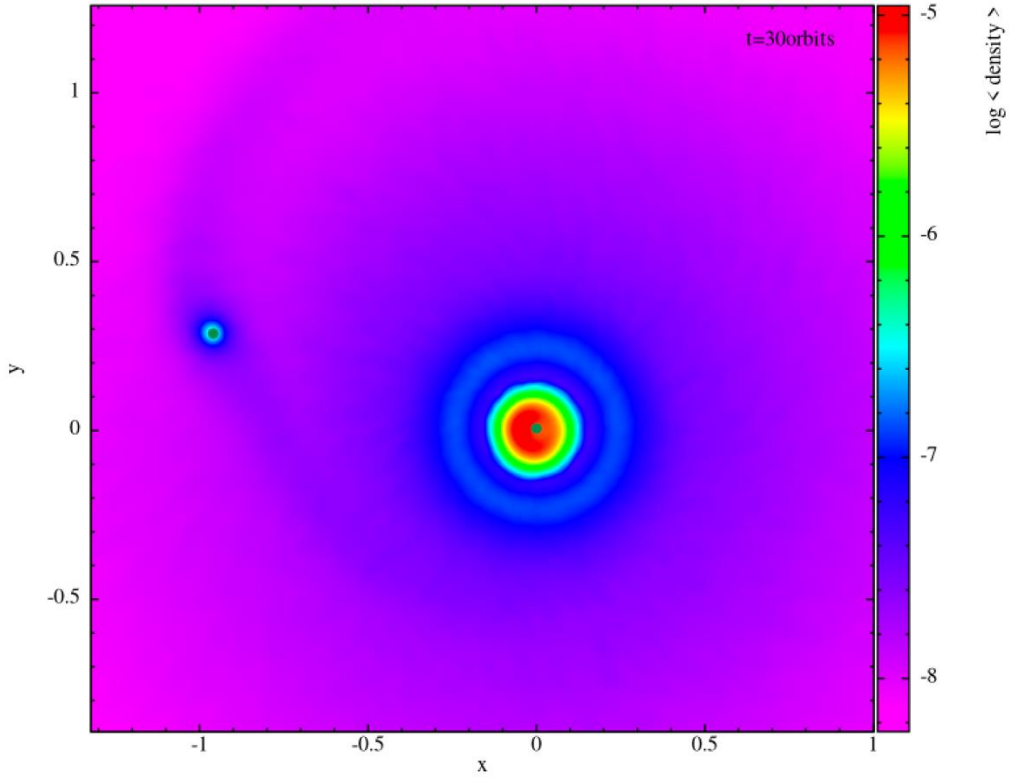
$n_x = 64. N_{smooth} = 64$ Reconstruction AV

Results of using spin algorithm

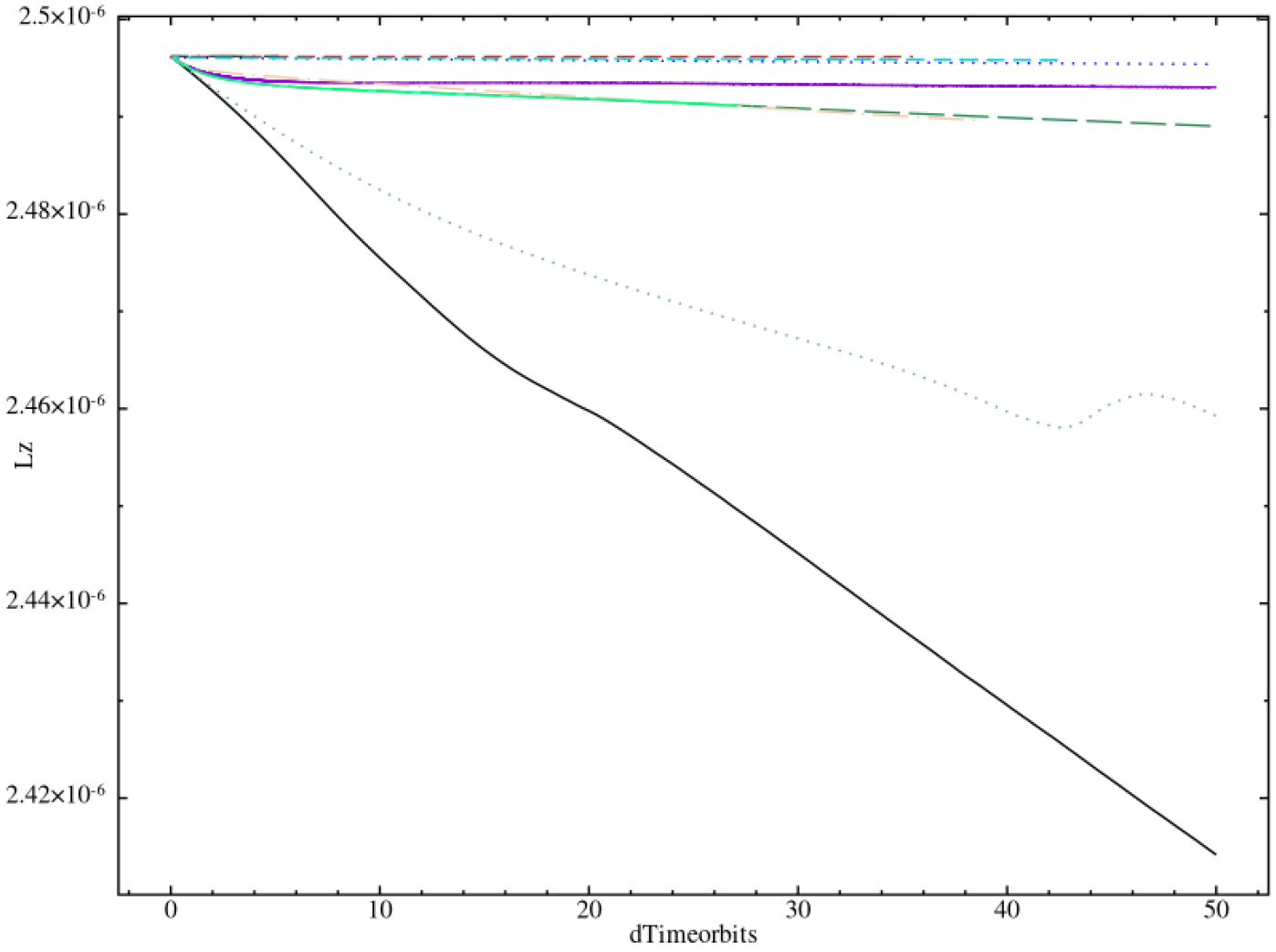
Spin



No Spin



Results of using spin algorithm



Spin

No Spin

Optimized smoothing kernels for SPH

- New smoothing kernel through a linear combination of kernels.

$$W_{new,i}(\alpha_i, \alpha_{i+1}) = (\alpha_i W_{new,i-1} + (1-\alpha_i)W_i)\,\alpha_{i+1} + (1-\alpha_{i+1})W_{i+1}$$

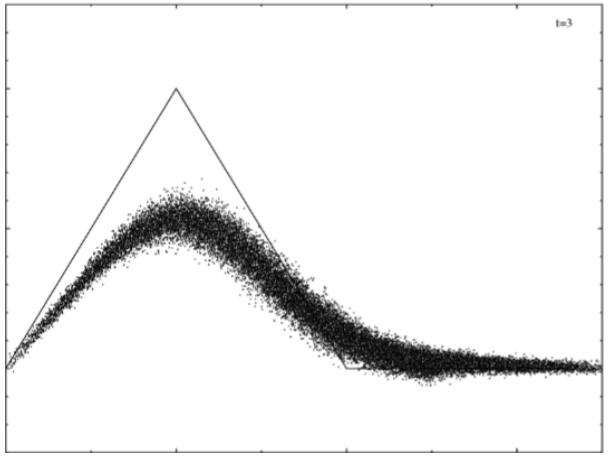
- After determining α_i, α_{i+1} by minimizing a cost function, we generate a new kernel.
- This iterative process continues until all desired kernels are included (12 kernels in this case)

- Cost function determined by running Gresho-Chan vortex simulation to t=1 and looking at the L1 error from the analytical solution

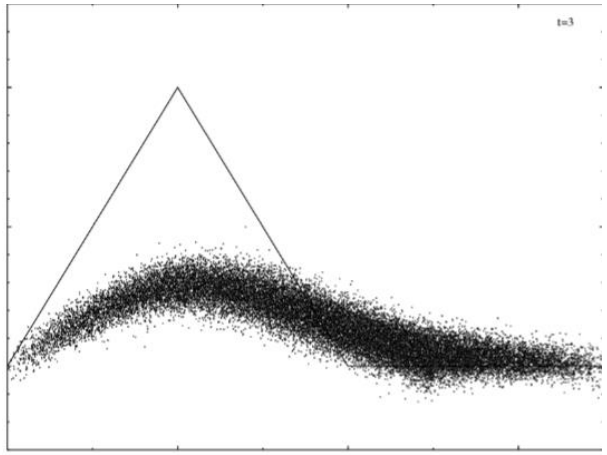
- We run this for both GDSPH and ISPH and for an array of neighbour numbers (32,64,128,256).

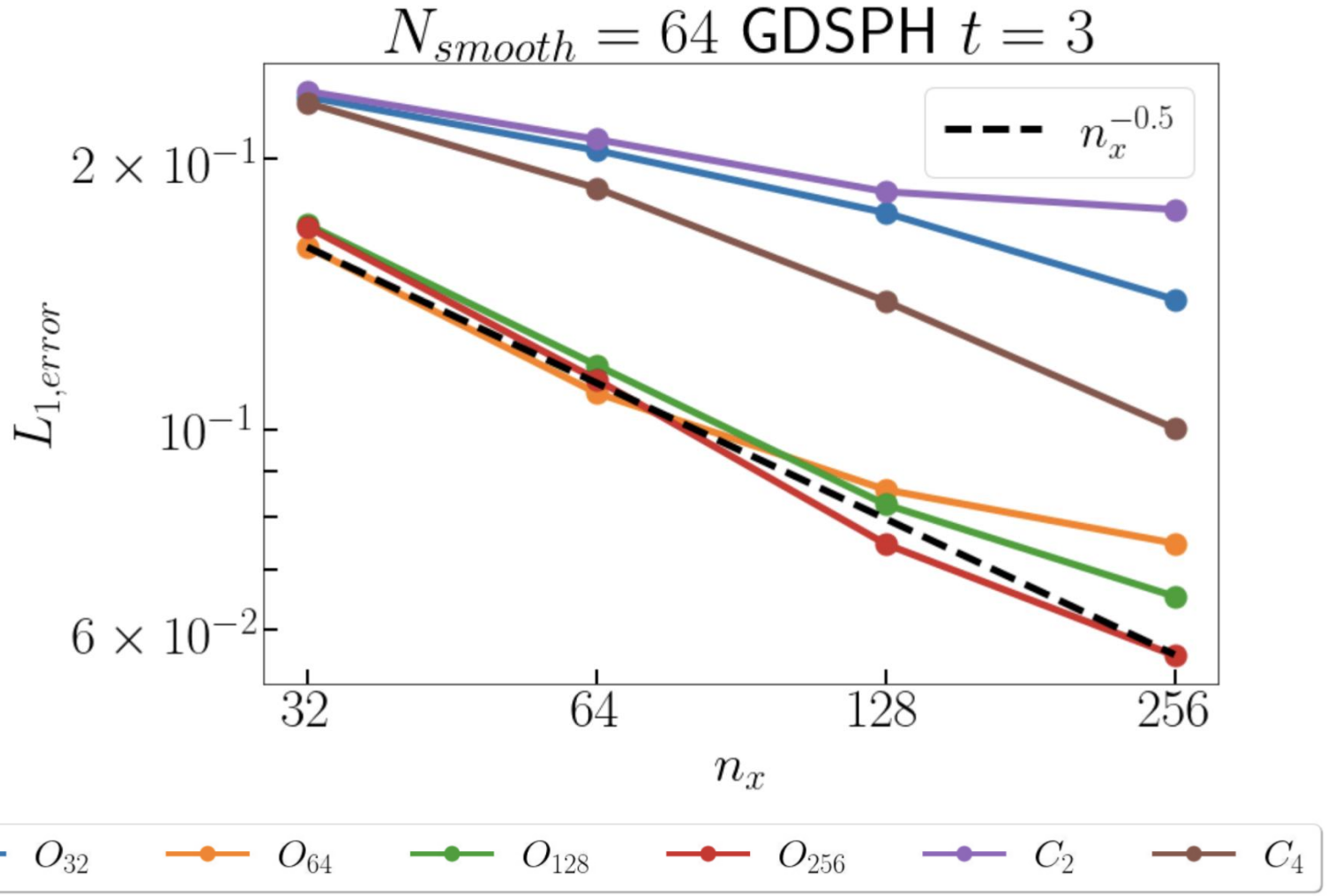
Optimized smoothing kernels for GDSPH

O_{64}



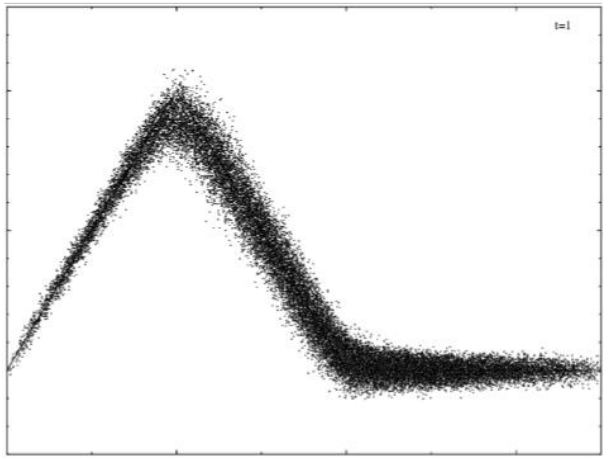
Wendland C4



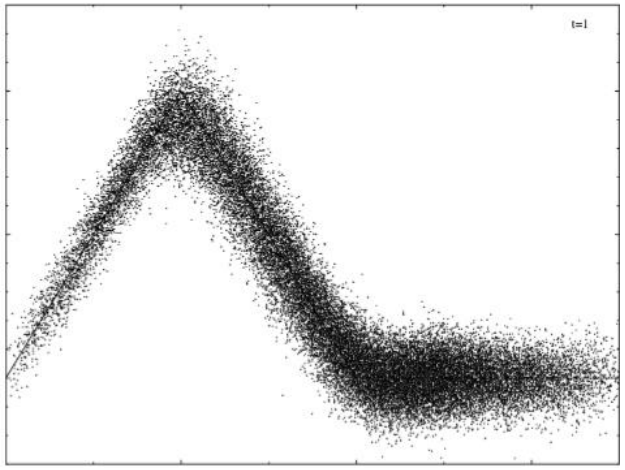


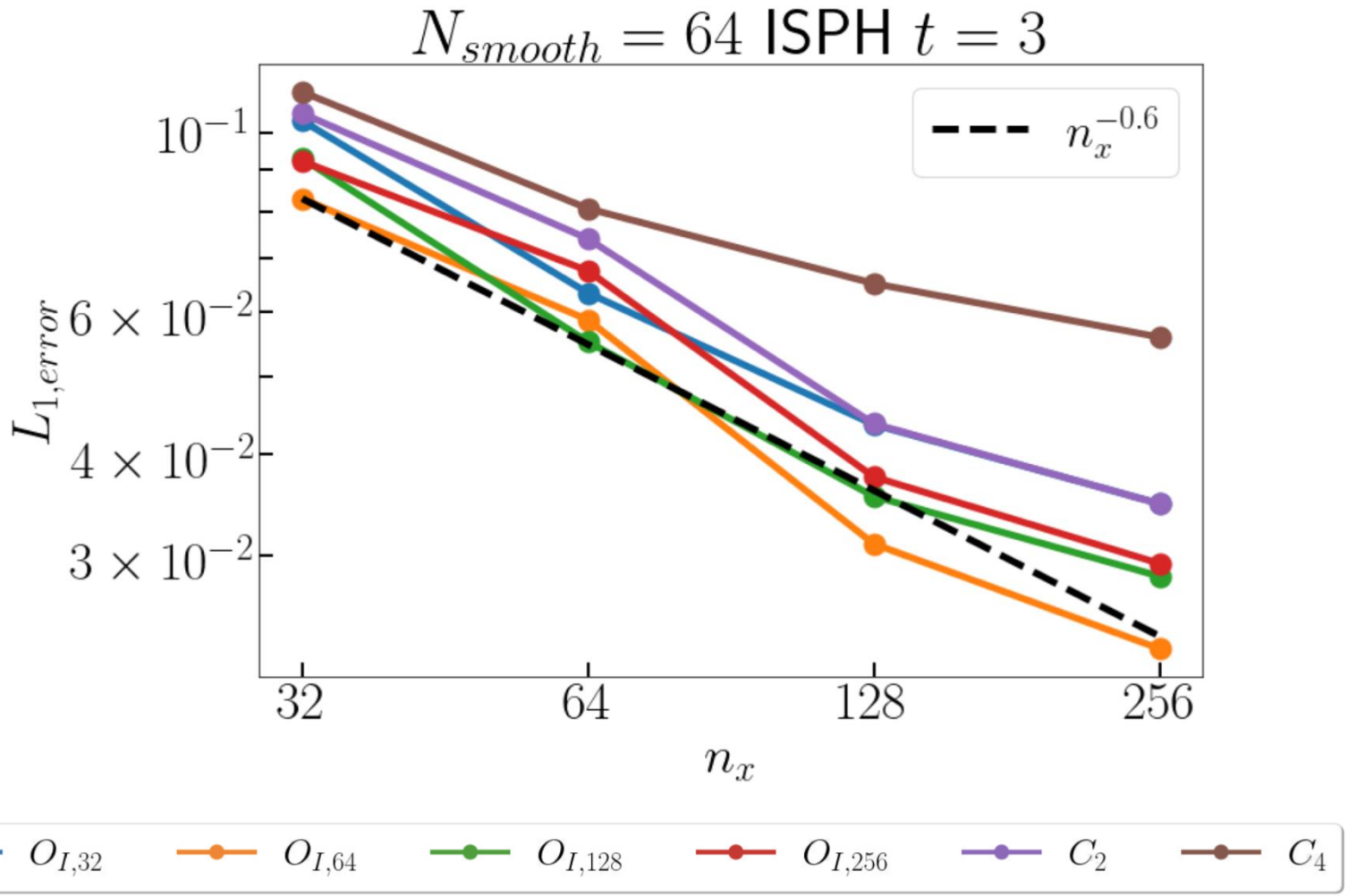
Optimized smoothing kernels for ISPH

$O_{I,64}$



Wendland C4





Optimized smoothing kernels for SPH

Optimized smoothing kernels for SPH

Robert Wissing, Thomas Quinn, Ben Keller , James Wadsley, Sijing Shen

https://arxiv.org/abs/2508.08471

Other improvements not mentioned in this talk:

Velocity reconstruction for dissipation (Frontiere et al. 2017)

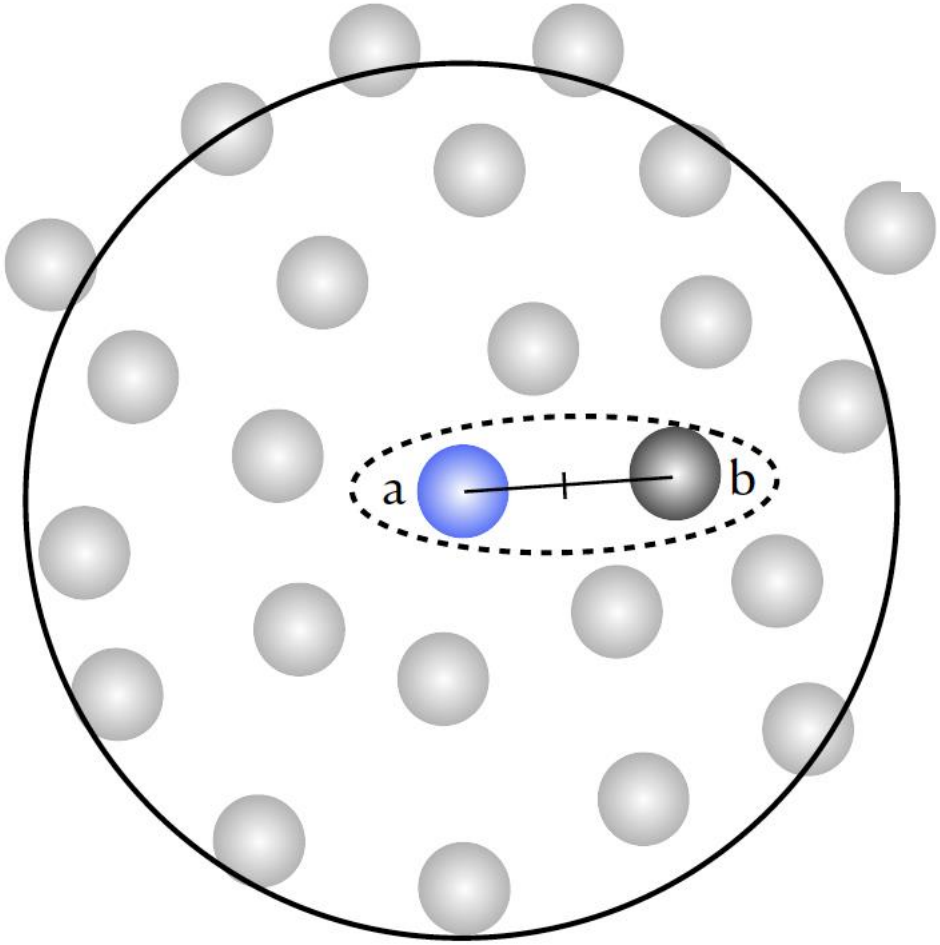
Improved volume partitioning in SPH

Acknowledgement & Disclaimer

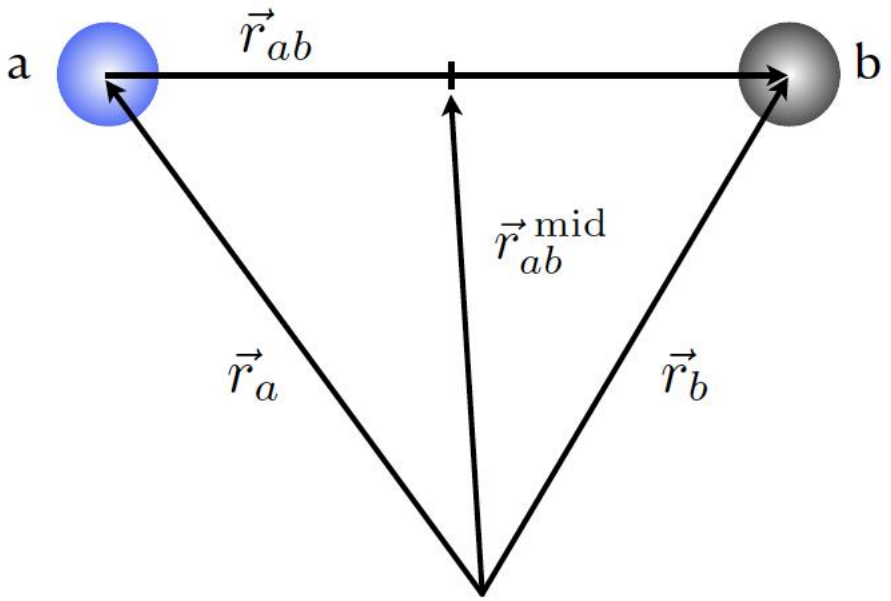
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Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European High Performance Computing Joint Undertaking (JU) and Belgium, Czech Republic, France, Germany, Greece, Italy, Norway, and Spain. Neither the European Union nor the granting authority can be held responsible for them

Reconstruction in SPH (Frontiere et al. 2018)



$$\tilde{v}_a^i = v_a^i + \Phi_{ab}\left[(\partial_j v^i)\,\delta^j + \frac{1}{2}(\partial_l \partial_m v^i)\,\delta^l \delta^m\right]_a,$$



Rosswog 2025

Reconstruction in SPH (Frontiere et al. 2018)

$$\tilde{v}_a^i = v_a^i + \Phi_{ab}\left[(\partial_j v^i)\,\delta^j + \frac{1}{2}(\partial_l \partial_m v^i)\,\delta^l \delta^m\right]_a,$$

$$\left(\frac{dv_a}{dt}\right)_{diss} = \sum_b m_b \Pi_{ab} \overline{\nabla_a W_{ab}},$$

$$\left(\frac{du}{dt}\right)_{diss} = -\sum_b m_b \left(\frac{\Pi_{ab}}{2}\right) v_{ab} \cdot \overline{\nabla_a W_{ab}},$$

$$\Pi_{ab} = \frac{-\alpha \frac{1}{2}\left(v_{sig,a} + v_{sig,b}\right)\mu_{ab} + \beta \mu_{ab}^2}{\frac{1}{2}\left(\rho_a + \rho_b\right)}$$

$$\mu_{ab} = \begin{cases} \frac{(\mathbf{v}_{ab} \cdot \mathbf{r}_{ab})}{|r_{ab}|} & \text{for } \mathbf{v}_{ab} \cdot \mathbf{r}_{ab} < 0, \\ 0 & \text{otherwise,} \end{cases}$$